The Theory of the GPS Composite Clock

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Biography

Kenneth Brown joined IBM in 1963 in Cambridge, Mass. while still an undergraduate at Harvard, where he took his A.B. (1964) and M.S. (1968). He has designed guidance and control algorithms for space vehicles and surface ships, and numerous Kalman filters, mostly for radar and sonar tracking systems. In 1983, he joined IBM's GPS team in Maryland to assume lead responsibility for the GPS Kalman filter. He invented what is now called the Composite Clock in 1987, showing how a system of orbiting clocks could be ensembled by Kalman filter. He is presently a Senior Engineer/Scientist with IBM in Gaithersburg, Md.

Abstract:

In essence, GPS is a network of atomic clocks related by measured signal transit times, specifically receive times measured by ground station clocks minus transmit times measured by satellite clocks. Such measurements allow both orbit and clock estimation, but since clocks appear only in differences, any error common to all the clocks would remain unobservable. Early GPS designs therefore treated one of the clocks as a master, "correct" by definition and dropped from the problem. Although such artificial reliance on a particular clock was obviously undesirable, suggestions of clock ensembling were inhibited by a lack of theory. The combined orbit/clock problem clearly required a Kalman filter or the equivalent, and that seemed to demand a completely observable system.

This paper presents the solution (in use since June 1990) known as the GPS Composite Clock. Essentially it is a Kalman filter applied to the original orbit/clock problem, including all clocks, treating none as master. There are no explicit ensembling equations, and the system is *not* completely observable, but a series of new theorems shows that (a) the Kalman filter *implicitly* creates an ensemble time from all the clocks, and (b) the unobservability can be handled in any of several acceptable ways.

A concept of transparent variations is developed to show the role of unobservable dimensions, and permit a dynamic separation of covariance into unobservable and observable parts. The observable part is recognized as the covariance of the various clocks about an *implicit ensemble mean* of them all, and the unobservable part as the covariance of the implicit ensemble mean relative to an ideal clock.

To within the small error levels of the observable parts of the system, every corrected clock (physical clock corrected by its estimated error) behaves *as if controlled* by the implicit ensemble mean. Thus a corrected clock *represents* the implicit ensemble mean (to within such errors) and has the same long-term stability. For example, long term Allan variances are typically reduced, relative to stand-alone values, by a factor of the effective number of clocks.

§1 Summary and Introduction

In this paper, a clock system is a system of n independent clocks related by clock-to-clock difference measurements. Additional (non-clock) states with independent prediction dynamics, such as the orbital states in GPS, can be included and can enter into measurements in arbitrary ways as long as the clock states appear only in clock-to-clock differences.

A clock system in this sense involves a fundamental unobservability: among the n clocks in the system, only the equivalent of n-1 are separately observable from measurements. This is because any error common to all clocks has no affect on any measurements. Initial formulations for GPS avoided this problem by designating one clock as the master, "correct" by definition and thus dropped from the estimation problem. Under this approach, the system was completely observable, but system time depended entirely on the designated master clock and was therefore limited by its accuracy and reliability.

This paper develops instead a fundamental approach to clock system estimation in which *all* the clocks are estimated in a full Kalman-Bucy filter. It differs from earlier Kalman filter designs by including the unobservable degrees of freedom implied by estimating all the clocks. It differs from prior clock ensembling algorithms by offering a general estimation framework, embracing non-clock states as well as clock states, and thus accommodating the orbiting clocks of GPS.

§1 explores the estimation problem for such a general system of n clocks related by clock difference measurements. In particular, §1.1 defines clocks as linear random processes with (typically) two states each. Since real clocks are designed to approximate an ideal clock, one concept of errors - deviations of a real physical clock from an ideal clock - arises even before any estimation problems are introduced. To avoid confusion with estimation errors, such physical clock errors are called bias vectors because they tend to be slowly varying.

§1.2 defines clock difference measurements and establishes notations for physical clock states, clock bias vectors, and bias vector estimates. §1.3 introduces the pivotal concept of a *corrected clock* - a physical clock corrected by an estimated bias vector. If estimated biases are made generally available (as in GPS), then any user in contact with a particular clock has, in effect, access to the corrected clock.

Both for historical perspective and as a foundation for later results, §1.4 shows the relation between the reduced problem of n-1 clocks, relativized to a designated master as in earlier GPS formulations, and the full n-clock estimation problem. It derives the former from the latter by means of

a mapping.

§2 introduces the concept of *transparent variations* in filter state and covariance. This allows separate analysis of the unobservable dynamics in an estimation problem even if actual state variables are mixtures of observable and unobservable dimensions. Transparent variations are changes in filter state and covariance that are transparent to the Kalman filter, i.e., have no effect on measurement residuals or filter gains or dynamics. They represent the ways in which filter state or covariance can be adjusted while maintaining the conditions for optimal filtering.

Among the transparent or unobservable components of covariance for a clock system, it is shown that there is always a largest (in the semi-definite sense). This is called *the* unobservable component of covariance, and what remains upon removing it is called *the* observable component of covariance.

§3 succeeds in analyzing error growth for the corrected clocks in an optimal n-clock system. This is done by defining the *implicit ensemble mean* of the corrected clocks (§3.1), an intermediate quantity that separates the unobservable and observable parts of the error growth for the system. The implicit ensemble mean is not computed explicitly since it depends on all the physical clock states which are never available in any one place in systems like GPS. Nonetheless, every corrected clock in the system always *represents* the implicit ensemble mean as accurately as possible given the available measurements.

More specifically, the errors in the corrected clocks are shown to consist of two components with radically different behaviors and dependencies: one component, the error in the implicit ensemble mean, is common to all the clocks, unobservable, and grows slowly but ultimately without bound regardless of measurement data. The other, called the *representation error,* is specific to each clock, responds to measurement feedback and thus generally remains quite small. It is shown that these two components are statistically independent, and their respective covariances are identified as the unobservable and observable components of covariance previously described in §2.2.

The error in the implicit ensemble mean grows, without benefiting from measurements, much like the error in a single stand-alone clock but with much smaller variances, typically by a factor of n where n is the effective number of clocks. On the other hand, the representation errors, defined as differences between the corrected clocks and the implicit ensemble mean, are seen in §3.2 to arise from observable subsystems, depend strongly on available measurements, and have small error covariances or even a steady state (measurement and process noises permitting). Thus the long-term stability of a corrected clock becomes that of the implicit ensemble mean, typically n times better in Allan Variance than its stand-alone stability (§3.5).

Under fairly common conditions (the PPN case of §3.3), the implicit ensemble mean and its covariance are shown to be independent not only of the values of the measurements but also of their number and quality (variances) and even of their existence. For PPN cases, the implicit ensemble mean is equivalent to a weighted mean of the physical clocks with fixed weights, regardless of any measurements. [A contrasting case involving clock reinitializations, in which the existence of measurements is temporarily vital to keeping system time, is described in §3.4.]

Representation errors, in contrast, are always affected by measurements and generally kept small by them. Formulas are given in §3.2 for computing the actual covariance of the representation errors, that is, the observable component of covariance. To within representation errors, the filter bias estimates directly show individual physical clock behavior in relation to the implicit ensemble mean. This provides a method of monitoring individual clock performance.

§4 generalizes the results of §2 and §3 to sytems with arbitrary non-clock states, provided that the dynamics of the clocks states and non-clock states are separate, and provided that the clock states enter into the measurement equations only as clock to clock differences.

§5 considers ways of controlling covariance growth in a full n-clock system. In particular, the transparent methods described in §2 are compared with an alternative, now used in GPS, called a pseudo-measurement.

Most of the results in this paper appeared first in a contract report [5] submitted to the government.

§1.1 Clock Models

Usual state-space models for modern atomic clocks use either 2-state or 3-state polynomial processes driven by white noises. We will use the simpler and more common model, a two-state polynomial white noise process x(t) which may be written as follows:

$$(1)\qquad \frac{d}{dt}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}w_1(t)\\ w_2(t)\end{bmatrix}$$

or in vector form:

$$(2)\qquad \dot{x}(t) = Fx(t) + w(t) \qquad \text{where } F = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$$

where $w_1(t)$, $w_2(t)$ are independent white noises with variance rates (noise densities) $q_1(t)$, $q_2(t)$. That is,

$$(3)\qquad E\left[\int_t^{t+\tau} w_j(u)\,du\right]^2 = \int_t^{t+\tau} q_j(u)\,du \approx \tau\, q_j(t), \qquad j=1,2$$

where $E[\cdot]$ is the expectation operator. The state vector x(t) of the clock will be called the physical state of the clock.

The process model can be written equivalently in discrete time as follows:

$$(4)\qquad \begin{bmatrix}x_1(t_{k+1})\\ x_2(t_{k+1})\end{bmatrix} = \begin{bmatrix}1 & \tau\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1(t_k)\\ x_2(t_k)\end{bmatrix} + \begin{bmatrix}w_1(k)\\ w_2(k)\end{bmatrix}$$

or in vector form:

$$(5)\qquad x(t_k) = \phi(\tau)x(t_{k-1}) + w(k), \qquad \phi(\tau) = \begin{bmatrix}1 & \tau\\ 0 & 1\end{bmatrix}$$

where $\tau = t_k - t_{k-1}$, and the discrete time process noise covariance is:

$$(6)\qquad Q_k(\tau) = E[w(k)w(k)^T] = \begin{bmatrix} q_1\tau + \frac{q_2\tau^3}{3} & \frac{q_2\tau^2}{2} \\ \frac{q_2\tau^2}{2} & q_2\tau \end{bmatrix}$$

where we have assumed for that $q_1(t)$, $q_2(t)$ can be treated as constants q_1, q_2 at least over short intervals (t_k, t_{k+1}).

The justification of such clock models has been discussed extensively; see for example references [1], [2], and [3].

System of n Independent Clocks

Individual clocks in a system of clocks will be distinguished by superscripts. Thus the state at t_k of the i-th clock is the vector $x^i(t_k)$, the process noise for the k-th time interval is $w^i(k)$ and its covariance is $Q^i(k)$. The process noise sequences $w^i(k)$ and $w^j(k)$ for distinct clocks i and j are independent.

Equations for an entire system of n clocks can be written in terms of a super-vector X(t) consisting of all the n subvectors $x^i(t)$, i=1,...,n. For clarity, we illustrate equations for the case n = 4, a number large enough to make patterns visually obvious in the matrix equations. The results, however, are general, applying to any n > 1.

$$(7) \qquad x^i(t_{k+1}) = \phi(\tau)\, x^i(t_k) + w^i(k) \qquad i=1,\ldots,n,$$

$$(8) \qquad X(t_k) = \Phi(\tau)\, X(t_{k-1}) + W(k), \qquad \text{where}$$

$$X(t)=\begin{bmatrix} x^1(t) \\ x^2(t) \\ x^3(t) \\ x^4(t) \end{bmatrix} \quad \Phi(\tau)=\begin{bmatrix} \phi(\tau) & 0 & 0 & 0 \\ 0 & \phi(\tau) & 0 & 0 \\ 0 & 0 & \phi(\tau) & 0 \\ 0 & 0 & 0 & \phi(\tau) \end{bmatrix} \quad W(k)=\begin{bmatrix} w^1(k) \\ w^2(k) \\ w^3(k) \\ w^4(k) \end{bmatrix}$$

where the blocks in X(t) and W(k) are all 2-vectors, and the blocks in $\Phi(\tau)$ are all 2x2 submatrices. The fact that $\Phi(\tau)$ has a common diagonal block $\phi(\tau)$, that is, a prediction model common to all clocks, is central to the idea of clock relativization discussed below and in §1.4.

The equation for Kalman filter covariance prediction can likewise be written in terms of super-matrices:

$$(9) \qquad \tilde{C}(t_k) = \Phi(\tau)\, \hat{C}(t_{k-1})\, \Phi(\tau)^T + Q_k(\tau) \quad \text{where}$$

$$(10): \quad C=\begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \\ C_{41} & C_{42} & C_{43} & C_{44} \end{bmatrix}; \quad Q=\begin{bmatrix} Q^1(\tau) & 0 & 0 & 0 \\ 0 & Q^2(\tau) & 0 & 0 \\ 0 & 0 & Q^3(\tau) & 0 \\ 0 & 0 & 0 & Q^4(\tau) \end{bmatrix}$$

in which again each element represents a 2x2 submatrix. Here the tilde (~) indicates a covariance after a prediction operation, and the hat (^) indicates a covariance after a measurement update.

There is also additional structure; since C is symmetric, $C_{ij} = C_{ji}^T$, and there can be still more. For example, Q^i may be the same for all i if all clocks have equal quality.

A Theoretical Ideal Clock

A peculiar characteristic of clock systems is that the member clocks were all designed to approximate the behavior of an ideal clock. Thus clock errors in one sense - differences of actual clocks from a theoretical ideal clock - are of interest even before we introduce any estimation issues. To emphasize the distinction, the word "errors" will be reserved for estimation errors in the Kalman filter, and clock differences from an ideal clock are given instead a special name, *bias vectors*, because in systems like GPS they tend to change slowly.

An ideal clock is characterized by the fact that it has no process noise. We will define an ideal zeroeth clock with state vector $x^0(t)$, using the same dynamic model (7) except that its process noises are zero:

$$(11) \qquad q_1^0 = q_2^0 = 0. \quad \text{Hence}$$

$$(12) \qquad Q^0(\tau) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}; \quad w^0(k) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{for all } k$$

The absence of process noises makes clock zero a deterministic process relative to its initial values. However, apart from estimates based on access to the real clocks, the actual values of the states of the ideal clock are a priori unknown, so the initial values of the ideal clock are regarded as random variables with large (essentially infinite) variances.

The a priori information on the state of an ideal clock can therefore be represented by an initial condition for clock 0 with infinite uncertainty:

$$(13) \qquad x^0(0) = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} \quad \text{where } \sigma^2_{\epsilon_i} = \infty \quad \text{for } i=1,2$$

Given a definition of the ideal clock, we can then define the bias vector of each individual clock as the difference $b^i(t)$ between the physical state $x^i(t)$ of the i-th real clock and that of the ideal clock:

$$(14) \qquad b^i(t) = x^i(t) - x^0(t) \qquad \text{Def of } b^i(t)$$

In GPS, the two components of this bias vector are often called the bias and drift components of the i-th clock.

Since the process noise vector $w^0(k)$ for clock zero is always zero, the i-th bias vector obeys exactly the same process model as the i-th clock state $x^i(t)$, given in (7) above:

$$(15) \qquad b^i(t_k) = \phi(\tau)\, b^i(t_{k-1}) + w^i(k)$$

and the super-vector B(t) of bias vectors $b^i(t)$, i=1,...,n, also obeys the same model as the super-vector X(t) of physical clock states in (8):

$$(16) \qquad B(t_k) = \Phi(\tau)\, B(t_{k-1}) + W(k)$$

Given the foregoing, we can now see how the definition of the clock processes might well have been approached from a different direction. We might have defined the ideal clock process first using (11)-(13) and a special case (i=0) of (7), then separately defined the bias vector processes using (15). Then the physical clock processes could have been defined as deviations from the ideal clock, via another form of (14):

$$(17) \qquad x^i(t) = x^0(t) + b^i(t) \qquad \text{Def of } x^i(t)$$

from which the general cases of (7) and (8) would have followed, along with (9) and (10).

This alternative way of defining clock processes is in some respects more revealing. It treats individual physical

clocks as perturbations of an ideal clock process defined by an additive random process called the bias vector. This view reflects the idea, developed in later sections, that the essential role of physical clocks is to provide information on the state of an ideal clock.

Moreover, the alternative definition (17), generalized to super-vectors representing all n real clocks, involves a matrix notation fundamental to later sections:

$$(18) \quad X(t) = H^* x^0(t) + B(t) \quad \text{where}$$

$$(19) \quad H^* = \begin{bmatrix} I \\ I \\ \cdot \\ \cdot \\ I \end{bmatrix} \quad \text{where } I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \text{Def of } H^*$$

Fundamentally, H^* is simply a replication matrix. As (18) represents n instances of (17), one for each real clock, (19) consists of n submatrices all of which are 2x2 identity matrices. Later on, H^* will appear in various other roles as well, but all are based on the fundamental relation (18) between physical clocks, the ideal clock, and bias vectors.

§1.2 Clock Difference Measurements

A system of clocks as independent random processes is statistically coupled by the effects of measurements. The measurements are on differences between the first states of two clocks, $x_1^i(t) - x_1^j(t)$, for some $i \geq 1, j \geq 1$.

Nominally measurements are valid at ideal time points t_k. But actual measurements are triggered by events within individual clocks, and are therefore valid at times as determined by the physical clocks.

Validity times can be corrected, to within filter accuracy. But even uncorrected validity times are generally acceptable because clock differences are extremely slowly varying. If the largest error rate in either clock is $\dot{\epsilon}$ (a pure number, sec./sec.), then the error induced in each clock solely by a validity time error δ is in the range from $(1-\dot{\epsilon})\delta$ to $(1+\dot{\epsilon})\delta$, so the error induced in the clock-to-clock difference is at most $2|\dot{\epsilon}\delta|$ in magnitude.

Even on a wristwatch, $\dot{\epsilon}$ is around 10^{-4}, and typical atomic clocks are 5 to 10 orders of magnitude better. So validity time errors are attenuated by many orders of magnitude in their effect on difference measurements, justifying a truly negligible increase in the measurement noise budget.

In Kalman filter notation, a measurement z is generally represented as a linear combination of state components, plus noise,

$$z = H X + v \quad \text{where } v = \text{measurement noise}$$

where H is called the measurement coefficient matrix. In the present case, where each measurement is the difference of two clock readings, H has a very specific structure. As before, illustrating with four clocks,

$$(1) \quad H = [h^1 \ 0 \ 0 \ h^2 \ 0 \ 0 \ h^3 \ 0 \ 0 \ h^4 \ 0 \ 0]$$
$$h^k = \delta_{ik} - \delta_{jk} \text{ for all k, for some distinct i,j}$$

where δ represents the Kronecker delta function. In other words, for some distinct i and j, we have $h^i=1$, $h^j=-1$, and all other $h^k=0$.

Because of this special structure for H, a clock-to-clock difference measurement involves the first component of the ideal clock exactly once positively and once negatively, and thus turns out independent of it.

$$(2) \quad H H^* = 0$$
$$H X(t) = H B(t)$$

We have seen that the bias vector B of the system of clocks obeys exactly the same process and measurement equations as the original vector X of absolute clock states. Relativization to an ideal clock has no effect on the estimation problem structure. We therefore can and will define the filter problem as one of estimating the clock bias vectors $b^i(t)$ rather than the absolute states $x^i(t)$.

Accordingly, the filter estimates will be denoted by $\hat{b}^i(t)$ for clock i, or $\hat{B}(t)$ for the system as a whole. As usual, filter estimates are conditional means given the available measurements, in this case clock difference measurements. Usually the convention is followed that the hat (^) indicates a conditional mean based on measurements up to and including the current time, whereas the tilde (~) will indicate a conditional mean based on measurements up to but excluding the current measurement.

The errors in the estimates will be denoted by $e^i(t)$ for clock i, or E(t) overall, where

$$(3) \quad b^i(t) = \hat{b}^i(t) + e^i(t), \text{ for all i,}$$
$$B(t) = \hat{B}(t) + E(t)$$

The fact that any clock difference measurement is independent of the state of the ideal clock can be formalized and generalized somewhat. Based on the structure of the measurement coefficient matrix H as described in (1), we saw already that $H H^* = 0$, but a little algebra shows that this property generalizes to all other times as well:

$$(4) \quad H \Phi^k H^* = 0 \quad \text{for all k}$$

A comment is in order on time dependencies. In GPS, like most clock systems, the availability of measurements normally follows a repeatable pattern. Thus, despite occasional data dropouts, the average quality of measurements per unit time is then roughly constant at least over suitably large time intervals. For a completely observable system with constant dynamics, this would usually imply an approximate statistical steady steady state, at least as seen over large enough time intervals. This paper, however, considers systems that are not completely observable, but for which some completely observable subsystems exist (a fundamental example is given in §1.4.) In such cases, only the observable subsystems can approach a steady state.

§1.3 Physical Clocks and Corrected Clocks

It is always important to distinguish clearly among physical quantities, estimates, and errors, especially with clock systems. We decided in the last section to call differences between physical clocks and an ideal clock, *bias vectors* $b^i(t)$, reserving the word *error* for estimation errors. The filter maintains estimates $\hat{b}^i(t)$, that is, expectations of bias vectors $b^i(t)$, and the difference between $\hat{b}^i(t)$ and $b^i(t)$ is the filter estimation error $e^i(t)$.

As a result of these conventions, the expectation of the

state of the ideal clock, given the physical state of clock i is simply:

$$(1) \qquad \omega^i(t) = E[x^0(t) \mid x^i(t)] = x^i(t) - \hat{b}^i(t) \qquad \text{Def of } \omega^i$$

where, as before, the conditional expectation includes implied conditioning on all the measurements available to the filter.

This quantity $\omega^i(t)$ is called the *corrected* state of clock i because it consists of the physical clock state corrected by the filter's estimate of its difference from an ideal clock.

As estimates of the state of an ideal clock, corrected clocks are fundamentally important. A corrected clock can represent system time. A user receiving timing signals from a physical clock and applying corrections (estimated biases) determined by the Kalman filter, ends up with data corresponding to the corrected clock. Because such signals are corrupted by receiver noise and other effects, access to the corrected clock may be imperfect, but the concept of a corrected clock is nonetheless extremely relevant for users of a system such as GPS.

Throughout this paper, the concept of corrected clock states plays a central role along with two related concepts defined in §3: the implicit ensemble mean of a system (a weighted combination of the corrected clocks), and the clock representation errors (differences between corrected clocks and the implicit ensemble mean).

Such a breakdown is not gratuitous but essential. The errors in the corrected clocks, which characterize the system as perceived by any user, will be shown to consist inherently of two components with radically different behaviors: one component common to all clocks, growing with time, and unobservable, and the other unique to each clock, observable, and having a steady state.

§1.4 Relativization to a Master Clock.

For all its shortcomings, the master clock concept used in early GPS designs has a much sounder basis than is usually given for it. The usual basis is naive: simply pretend the master clock is deterministic (no initial covariance and no process noise), hence drop it from the set of estimated states, and give equations for estimating the remaining n-1 clocks. This amounts to treating a real clock as if it were an ideal clock. There is then no attempt to relate the resulting design to the original n-clock problem.

In this section we deal instead with the full n-clock problem and show how to derive without any pretenses a fully optimal estimation algorithm for n-1 relativized clocks. In the process, we will see the reason for the clock-to-clock correlated process noise used in early releases of GPS to allow for perturbations in the unestimated master clock.

As we defined a relativization to an ideal clock in §1.1, we can also define a relativization to any one of the real clocks of the system. The results, however, are not the same; real clocks are random processes with non-zero process noises.

For illustration, we will relativize all clocks to clock 1 by a state transformation that subtracts the state of clock 1, $x^1(t)$, from the state $x^i(t)$ of every clock except clock 1 itself, which is unchanged. The relativized states will be denoted by primes, e.g. $x'^i(t)$ for each clock i and $X'(t)$ for the system as a whole:

$$(1) \qquad X'(t) = M\,X(t) \qquad \text{where}$$

$$(2) \qquad M = \begin{bmatrix} I & 0 & 0 & 0 \\ -I & I & 0 & 0 \\ -I & 0 & I & 0 \\ -I & 0 & 0 & I \end{bmatrix} \qquad M^{-1} = \begin{bmatrix} I & 0 & 0 & 0 \\ I & I & 0 & 0 \\ I & 0 & I & 0 \\ I & 0 & 0 & I \end{bmatrix}$$

Due to the block diagonal structure of $\Phi(\tau)$ and the fact that the diagonal blocks are all equal to $\phi(\tau)$, the block mapping M preserves $\Phi(\tau)$:

$$(3) \qquad \Phi'(\tau) = M\,\Phi(\tau)\,M^{-1} = \Phi(\tau)$$

But the process noise matrix $Q'(\tau)$ for X' is different from that for X:

(4):

$$Q' = M\,Q\,M^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & Q^2 & 0 & 0 \\ 0 & 0 & Q^3 & 0 \\ 0 & 0 & 0 & Q^4 \end{bmatrix} + \begin{bmatrix} Q^1 & -Q^1 & -Q^1 & -Q^1 \\ -Q^1 & Q^1 & Q^1 & Q^1 \\ -Q^1 & Q^1 & Q^1 & Q^1 \\ -Q^1 & Q^1 & Q^1 & Q^1 \end{bmatrix}$$

where for compactness we have omitted the arguments (τ) throughout. This would agree with the original matrix $Q(\tau)$ if $Q^1(\tau)$ were zero, that is, if clock 1 were perfect, which is the assumption of the naive approach to a master clock system design.

But even with a realistic clock 1 with a non-zero $Q^1(\tau)$, the mapping M can still be applied successfully to the whole filtering problem, not just the prediction part, to produce a reduced-dimension estimation algorithm encompassing the n-1 relativized clocks alone.

To see this, first consider what becomes of the measurement matrices H under the mapping M. Since each measurement originally represents a difference between the readings of two clocks, say clocks i and j, there are only two possibilities. Case 1: Both i and j are > 1. In this case the subtraction of clock 1 from both has no effect on the difference, so the measurement has the same definition after the mapping as it did before. Case 2: One of the two clocks is clock 1. In this case, since the mapping already computes the difference between the other clock and clock 1, the measurement actually equals (plus or minus) the 1st component of the other relativized clock. So in both cases the measurement, considered as a function of X' rather than X, is a function of clocks 2' through 4' only, independent of clock 1'. This is represented by the following equation:

$$(5) \qquad H' = H\,M^{-1} = [\,0 \quad H'_2 \quad H'_3 \quad H'_4\,]$$

in which, assuming we consider measurements one at a time, H and H' are 1x8 matrices and the zero and non-zero submatrices are each 1x2.

The first submatrix of H' is the only one that always vanishes. In fact, for any j other than 1, there is a clock difference measurement such that $H'_j = (1\ 0)$ and the rest of H' is zero.

Next we will show that the vanishing of the first submatrix of H' has the consequence that the lower right (n-1)x(n-1) array of covariance blocks corresponding to the n-1 relativized clocks can be propagated by itself without reference to the first row or column of blocks. This is obvious for the prediction part of covariance propagation

since Φ' is block diagonal. The correction part, using standard filter equations, is

$$(6) \qquad \hat{C} = \tilde{C} - \tilde{C}\,H'^{T}\left[\,R + H'\tilde{C}\,H'^{T}\right]^{-1} H'\tilde{C}$$

We can repartition H' and C as follows:

$$(7) \qquad C = \left[\begin{array}{c:c} A & B \\ \hdashline B^{T} & D \end{array}\right] \qquad H'^{T} = \left[\begin{array}{c} 0 \\ \hdashline J^{T} \end{array}\right]$$

where A is the covariance of clock 1' alone and D is the much larger covariance of the remaining n-1 relativized clocks taken as a group. Using this partitioning to evaluate the measurement update for covariance, we have

(8):

$$\hat{C} = \left[\begin{array}{c:c} \hat{A} & \hat{B} \\ \hdashline \hat{B}^{T} & \hat{D} \end{array}\right] = \left[\begin{array}{c:c} \tilde{A} - \tilde{B}\,J^{T}V\,J\,\tilde{B}^{T} & \tilde{B} - \tilde{B}\,J^{T}V\,J\,\tilde{D}^{T} \\ \hdashline \tilde{B}^{T} - \tilde{D}\,J^{T}V\,J\,\tilde{B}^{T} & \tilde{D} - \tilde{D}\,J^{T}V\,J\,\tilde{D}^{T} \end{array}\right]$$

where V stands for the weighting matrix:

$$(9) \qquad V = \left[\,R + H\,\tilde{C}\,H^{T}\right]^{-1} = \left[\,R + J\,\tilde{D}\,J^{T}\right]^{-1}$$

The lower right partition of (8), together with (9), shows that block D, representing the relativized clocks 2' through 4', can be updated without any knowledge of the other parts (A and B) of covariance.

Recapitulating, we have shown that the relativizing transformation represented by the matrix M above, which replaces each clock (except clock 1) by its difference from clock 1, gives rise to optimal filter equations for which the lower right (n-1)x(n-1) array D of blocks representing covariance of the relativized clocks, can be propagated by itself as follows:

(10)

$$\tilde{D} = \begin{bmatrix} \phi & 0 & 0 \\ 0 & \phi & 0 \\ 0 & 0 & \phi \end{bmatrix} \hat{D} \begin{bmatrix} \phi & 0 & 0 \\ 0 & \phi & 0 \\ 0 & 0 & \phi \end{bmatrix}^{T} + \begin{bmatrix} Q^{2} & 0 & 0 \\ 0 & Q^{3} & 0 \\ 0 & 0 & Q^{4} \end{bmatrix} + \begin{bmatrix} Q^{1} & Q^{1} & Q^{1} \\ Q^{1} & Q^{1} & Q^{1} \\ Q^{1} & Q^{1} & Q^{1} \end{bmatrix}$$

$$(11) \qquad \hat{D} = \tilde{D} - \tilde{D}\,J^{T}\left[\,R + J\,\tilde{D}\,J^{T}\,\right]^{-1} J\,\tilde{D}^{T}$$

where J is as defined by (7) above. This proves the following:

Theorem 1.4: Clock System Relativization

Given a system of n independent clocks with measurements on clock-to-clock differences, a subsytem of n-1 relativized clocks (states differenced from a particular clock) can be optimally estimated by an (n-1)-clock Kalman filter with suitable correlated process noise. The relativized subsystem is completely observable from clock difference measurements alone.

This is a fully optimal basis for estimation of the n-1 non-master clocks in relation to a designated master clock. This refinement was first deployed in GPS in 1984 and remained till June, 1990 when replaced by the method of implicit ensembling presented below. Apart from its improved soundness of theoretical foundation, it differed from the naive approach by recognizing the added, clock-to-clock correlated, process noise blocks $Q^{1}(\tau)$ induced in the relativized clocks by the randomness of the unestimated master clock.

Although important, this refinement of the master clock concept did not solve or even address its most basic weaknesses. The stability and reliability of system time remained limited by the master clock. The operational problems, such as master clock failure modes and reassignment issues, remained as before.

There was a performance advantage of the revised master clock formulation. In restoring the missing process noise blocks $Q^{1}(\tau)$, it removed a basic filter inconsistency inherent in the naive approach. This improved relative errors generally but only slightly, of the order of 5-10%. The main effect was to allow the non-master clocks to adapt more uniformly and thus more quickly to disturbances in the master as shown by difference measurements between it and the rest.

The existence of an exact relativization mapping from the full n-clock estimation problem to a reduced-dimension problem, corresponding to the revised master clock formulation, implies two main results: (a) that there are many subsystems of the full n-clock problem that are completely observable, implying small variances and at least roughly steady-state statistics within each, and (b) clock-to-clock difference estimates and their covariances are *exactly* the same in the full n-clock system as in the revised master clock formulation.

§2 Transparent Variations

Discussions of process estimation algorithms often begin with a definition of complete observability and then assume this property for problems of interest. However, in clock systems as defined in this paper, some dimensions of the state space are permanently unobservable.

For such systems, this section introduces a concept of transparent variations - changes in either state estimates or covariances that are transparent to an optimal filter in the sense that gains, measurement residuals, and filter performance are entirely unaffected. This approach lays a foundation for understanding the separate error growth characteristics for the observable and unobservable components of a problem. In addition, it shows ways in which filter states and covariance can be adjusted to meet requirements external to the estimation problem.

Kalman showed in [4] that a linear system can be decomposed into unobservable and observable parts in a way that essentially uniquely defines the dynamics and measurements of the observable part and permits it to be propagated independently of the rest. This might seem to imply that nothing is lost by simply doing so. In a clock system, however, the most vital information - time itself - lies in the unobservable space. The fact that one can do justice to the observable space, using any of a variety of reduced dimension filters that are equivalent within that space, is therefore not enough.

In fact, different reduced dimension filter designs for the

observable space are *not* equivalent with respect to the unobservable space. This is shown by the clock relativization results of §1.4, whose examples include different choices of master clock and thus different definitions of system time.

We therefore approach the subject differently, analyzing the role of unobservable dimensions in context - without assuming any prior transformation separating them from the observable dimensions. This paves the way for variable rather than fixed decompositions, leading eventually to a covariance-dependent definition of the unobservable and observable components of a problem, a definition that depends on the statistics of the system but not on the coordinate system in which it is defined.

§2.1 Unobservable Variations in State

A change δx_0 in state x_0 at time t_0 will be called unobservable if it has no effect on measurements at any time. This means not only that partial derivatives of measurements at t_0 with respect to x_0 have no component in the direction δx_0, but also that partials of measurements at any other time t with respect to states x(t) will have no component in the direction $\delta x(t) = \Phi(t,t_0)\delta x_0$ where $\Phi(t,t_0)$ is the state transition matrix.

For a clock system with measurements on clock-to-clock differences, an example of such an unobservable change δx_0 is a vector of equal offsets in the first components of all the initial clock states, with zeroes for all other state components (including non-clock states if any). Any difference measurement between two clocks at any time will involve the offset once positively and once negatively and so be unaffected.

If all clocks have two or more states, including a bias and a drift, then a vector δx_0 of equal offsets to all the drift states and zeroes for other states would also be unobservable. Similarly for drift rates if all clocks have three or more states.

So far we have referred to changes δx_0 in x_0 at starting times t_0, but any such implies an entire history of changes $\delta x(t) = \Phi(t,t_0)\delta x_0$ for other times t, each of which meets the same requirement for an unobservable change at time t. Such a history $\delta x(t)$ is simply a first variation of the state history x(t). It is simpler to think in terms of unobservable variations $\delta x(t)$ as functions of time than in terms of changes δx_0 at arbitrary starting points t_0.

Unobservable variations are most easily understood by focusing first on the continuous-time idealization of the estimation problem. Let the state x(t) of a system be a column vector of length n satisfying dynamic and measurement equations of the form

$$(1) \quad \dot{x}(t) = F(t)\, x(t) + w(t)$$
$$(2) \quad z(t) = H(t)\, x(t) + v(t)$$

where w(t), v(t) are process and measurement noises.

Definition 2.1: Unobservable Variation:

Given a dynamical and measurement system as in (1) and (2) above, an Unobservable Variation is a function of time $\delta x(t)$ such that for all t,

$$(i) \quad \dot{\delta x}(t) \equiv F(t)\, \delta x(t)$$
$$(ii) \quad H(t)\, \delta x(t) \equiv 0$$

The first condition states that $\delta x(t)$ as a function of time is in fact a first variation of the dynamic equations; the second, that it has no effect on measurements at any time.

In this definition, there need be no restriction that $\delta x(t)$ be merely an n-vector. It can be an nxm matrix, in which each of m columns may represent a different unobservable variation. In the case of a system of clocks with three states each, there could be three essentially different variation vectors that are unobservable, so it can be important to let m be as large as three. For GPS, m=2 is sufficient.

Under this definition, if an nxm matrix $\delta x(t)$ qualifies as an unobservable variation, then so does $\delta x(t)A$, where A is an arbitrary constant mxp matrix. In other words, a linear combination of unobservable variations is itself an unobservable variation.

The state elements recognized in a particular formulation place a limit on the number of independent degrees of freedom in the unobservable space for that formulation. We will consider an unobservable variation matrix $\delta x(t)$ to be *complete* if any unobservable variation of the estimation problem can be expressed as $\delta x(t)A$ for some A. It will also be called *minimal* if it ceases to be complete when deprived of any one column. Even a minimal complete unobservable variation $\delta x(t)$ is not unique, for it can be expressed in many ways. However, any two such, $\delta x_1(t)$ and $\delta x_2(t)$, must be mutually representable in terms of each other, e.g., $\delta x_2(t) = \delta x_1(t)A$ and $\delta x_1(t) = \delta x_2(t)A^{-1}$ for some matrix A.

For a system of clocks with two states each, it is easily seen that a minimal complete unobservable variation will have just two columns. The simplest example has in effect already been described above. Its initial value at some time has for the first column equal elements (e.g., ones) for the first components of all clocks and zeroes elsewhere. For the second column, it has ones for the second component of each clock and zeroes elsewhere. In short, H* as defined in §1.1, as a column of 2×2 identity matrices, is a possible value of an unobservable variation matrix that is both complete and minimal.

Of course, over time, the initial value H* propagates into various matrices of the form $\Phi(\tau)$ H* for various τ, each of which gives a different minimal complete unobservable variation matrix for the same problem.

Any unobservable variation, complete or not, remains apart from the feedback dynamics of optimal filtering, neither affecting nor being affected by the rest of the estimation problem. In a filter implementation, if the estimate $\hat{x}(t)$ is perturbed by a vector $\delta x(t)$ that satisfies the conditions for an unobservable variation, the perturbation simply remains in place forever. No measurement residuals are ever produced by such a change, so there is no dynamic response. Thus unobservable variations represent state changes transparent to the Kalman filter - changes that can be added without affecting filter dynamics.

§2.2 Unobservable Components of Covariance

Since most algorithm formulations result from first separating out and then eliminating any unobservable dimensions, the presence of any unobservable variations is

unusual in estimation algorithms. One may well ask what atypical behaviors for covariance will result if unobservable variations exist for state. One obvious effect, assuming the dynamics are fully controllable by process noise, is that filter covariance will grow without bound in some directions. But we can be much more specific:

Theorem 2.2: Transparent Variations of Covariance

If $\delta x(t)$ is an unobservable variation of an estimation problem as above, and if $K(t)$ and $C(t)$ satisfy the equations for the optimal filter gain matrix and covariance matrix, then so do $K(t)$ and $C'(t)$ given by

$$(iii) \qquad C'(t) = C(t) + \delta x(t)\ \delta x(t)^T$$

Optimal filter covariances C(t) are solutions of a quadratic differential equation, the matrix Riccati equation of the form:

$$(1) \qquad \dot{C} = F\,C + C\,F^T + Q - C\,H^T R^{-1} H\,C$$
$$= F\,C + C\,F^T + Q - K\,H\,C, \qquad \text{where}$$
$$(2) \qquad K(t) = C(t)\,H(t)^T R(t)^{-1} \qquad \text{(Kalman gains)}$$

Normally a quadratic differential equation couldn't allow superposition of linear homogeneous solutions or any others. But when unobservable variations exist, the quadratic term has a null space within which it does not respond, and this allows an additive freedom in covariance:

Proof:

First observe that if $\delta x(t)$ is an unobservable variation, then adding a covariance increment $\delta x(t)\delta x(t)^T$ does not affect the gain matrix:

$$(3) \qquad K'(t) = [C(t) + \delta x(t)\delta x(t)^T]H(t)^T R(t)^{-1} = K(t) + 0$$

since $H(t)\delta x(t)$ is identically zero by part (ii) of the definition.

Then evaluate the time derivative of $C'(t)$ based on the differential equation for optimal filter covariance. Omitting the time arguments,

$$(4) \qquad \frac{d}{dt}C' = F[C + \delta\,\delta^T] + [C + \delta\,\delta^T]F^T + Q - K'H\,C'$$
$$= \underbrace{FC + CF^T + Q - KHC}_{\frac{d}{dt}C} + \underbrace{F\delta\delta^T + \delta\delta^T F^T}_{\frac{d}{dt}[\,\delta\,\delta^T\,]}$$

Here the first step depends on the identity $K'=K$ and on part (ii) of the definition to justify replacing $K'H\,C'$ by $K\,H\,C$, and the last step uses part (i) of the definition to express the time derivative of $\delta(t)\delta(t)^T$. Thus $C'(t)$ as defined satisfies the equation for optimal filter covariance, q.e.d.

An obvious analogous result applies to discrete time formulations.

Covariance Decomposition

One consequence of this theorem is that when any covariance can be represented as a sum of another covariance $C(t)$ and $\delta x(t)\delta x(t)^T$, where $\delta x(t)$ is an unobservable variation of state, the $C(t)$ part propagates as a solution to the full covariance equation, while the other part propagates by pure prediction, unaffected by either measurement or process noise. Often covariances can be represented as such a sum in many different ways, and this idea can be applied repeatedly.

For example, in a stationary discrete-time problem, suppose the process noise increment matrix Q_k is representable as a sum: $Q_k = S_k + y_k y_k^T$ where S_k is a legal covariance (symmetric semidefinite) matrix and y_k is a possible value of an unobservable variation $\delta x(t)$. By repeated use of the theorem, the covariance at any time t_n can be expressed as the solution for filter covariance assuming $Q_k = S_k$ alone, plus the sum

$$\sum_{k \le n} \Phi^{n-k} y_k [\Phi^{n-k} y_k]^T$$

This shows explicitly how the unobservable part of the process noise propagates by pure prediction, unaffected by either measurements or the other components of process noise.

Unobservable and Observable Components

Whenever a covariance C can be expressed as $C = C' + \delta\,A\,A^T\delta^T$, where δ is an unobservable variation and C' is a possible covariance (i.e., C' is at least positive semidefinite), $\delta AA^T\delta^T$ will be called *an* unobservable component of C.

For a given covariance C, if unobservable variations δ exist at all there are generally many possible unobservable components of C, and for most the difference C' still has unobservable components in turn. We will call a decomposition $C = C' + \delta\,A\,A^T\delta^T$ *maximal* just in case the matrix C' has no (non-zero) unobservable components.

We can construct a maximal decomposition in this sense. Given a covariance C, every possible decomposition into observable and unobservable parts is given by subtracting a general matrix of the form $\delta\,A\,A^T\delta^T$ *provided that it is not too large;* otherwise the difference

$$(5) \qquad C' = C - \delta\,A\,A^T\delta^T$$

would not be semidefinite, thus failing to be a possible covariance.

We need to characterize the cases for which C' remains semidefinite. The set of semidefinite matrix differences $C' = C - S$, for symmetric semidefinite S, can be identified with the set of possible measurement updates of C for *arbitrary* possible measurements (not necessarily clock difference measurements). Given the measurement coefficient matrix H for such an arbitrary measurement (not necessarily a clock difference measurement) and given a corresponding measurement covariance matrix R, a measurement update would replace covariance C by C' as follows:

$$(6) \qquad C' = C - C\,H^T[R + H\,C\,H^T]^{-1} H\,C$$

The subtracted term in (6) can be of the form $\delta\,A\,A^T\delta^T$ only if

$$(7) \qquad C\,H^T = \delta\,Y, \qquad \text{for some matrix } Y$$

in which case,

$$(8) \qquad C' = C - \delta\,Y\,[R + Y^T\delta^T C^{-1}\delta\,Y]^{-1} Y^T\delta^T$$
$$= C - \delta\,[Y^{-1T} R\,Y^{-1} + \delta^T C^{-1}\delta]^{-1}\delta^T$$

which is indeed of the desired form $C - \delta\,A\,A^T\delta^T$ for some A. Moreover, for any given value of Y, the resulting C' is clearly minimized in the semidefinite sense by choosing R

to be zero, in which case C' becomes independent of Y, and thus minimal overall:

$$(9) \qquad C' = C - \delta\,[\delta^T C^{-1}\delta]^{-1}\delta^T$$

Thus (9) expresses the smallest semidefinite matrix C' such that

$$C = C' + \delta\, A\, A^T\delta^T, \quad \text{for some } A$$

It is easily shown that C' as given by (9) cannot be further decomposed relative to δ. That is, there can be no Y and H^T such that $C'H^T = \delta Y \neq 0$. To see this, observe that if there were, we would then have

$$\delta Y = C'H^T = C\,H^T - \delta\,[\delta^T C^{-1}\delta]^{-1}\delta^T H^T$$

Hence, pre-multiplying both sides by $Y^T\delta^T C^{-1}$, we would have

$$Y^T\delta^T C^{-1}\delta\, Y = Y^T\delta^T C^{-1} C\, H^T - Y^T\delta^T C^{-1}\delta[\delta^T C^{-1}\delta]^{-1}\delta^T H^T$$
$$= Y^T\delta^T H^T - Y^T\delta^T H^T = 0$$

contrary to the assumption that C was definite and $\delta Y \neq 0$.

If $\delta(t)$ is a complete unobservable variation matrix as defined in §2.1, that is, if *every* unobservable variation vector can be expressed as $\delta(t)A$ for some A, then C' as given by (9) can have no unobservable component at all, and thus defines a maximal decomposition of C.

It is clear from (9) that if δ_1 and δ_2 are two equivalent unobservable variations in that $\delta_1(t)=\delta_2(t)A$ for some $A\neq 0$, then they give the same maximal decomposition. Thus any two complete unobservable variations and for the same problem must give the same maximal decomposition.

Thus maximal decomposition is unique and justifies calling C' and C-C' *the* observable and *the* unobservable components of C.

In the case of a clock system as defined in §1, we can identify in advance a matrix, that is always a possible initial value of a complete unobservable variation matrix $\delta(t)$, namely the replication matrix H* defined in 1.1-(19), illustrating for the case of n = 4 clocks:

$$(10) \qquad H^* = \begin{bmatrix} I \\ I \\ I \\ I \end{bmatrix}$$

there being one 2x2 identity submatrix for each clock.

The first column of H* corresponds to a vector of equal offsets to the bias components of all clocks, and the second, equal offsets to the drift components. Note that this is only a possible starting value of an unobservable variation matrix. H* is not an unobservable variation as such because it does not vary in time as required to satisfy definition 2.1.

However, at any time t*, we can define a complete unobservable variation $\delta(t)$ whose value at t* is H*:

$$(11) \qquad \delta(t) = \Phi(t, t^*)H^*$$

So by (9), the unique maximal decomposition of C is:

$$(12) \qquad C' = C - H^*\,[H^{*T}C^{-1}H^*]^{-1}H^{*T}$$

in which C' is the observable component of C, and C-C' is the corresponding unobservable component.

We will meet the matrix H* and another version of equation (12) again in a very different context in §3: involving the implicit ensemble mean (there defined) and the accuracy with which individual corrected clocks represent that ensemble mean.

Control of Covariance Growth

For a clock system with no master, a central problem of Kalman filter design is the unwanted and ultimately unbounded growth of filter covariance. Covariance both determines and explains the relative weights of data over time and within the state space. For covariance to grow arbitrarily large seemed to threaten both the numerical stability and the meaning or interpretability of the filter.

The decomposition methods discussed above answer both these concerns. First, they let covariance be interpreted as the sum of an unobservable and an observable component. We will see in §3 that the former describes the slowly growing errors in an implicit ensemble mean of all the clocks, while the latter describes errors about that mean.

Indirectly, such decompositions answer the threat of numerical instability as well. To whatever extent desired, we are free to replace covariance at will using equation (8) for any Y, any arbitrary unobservable variation $\delta(t)$, and any positive definite R. The subsequent filter dynamic response will not be affected at all. An obvious choice is to replace covariance by its observable component C' as in (12) which, we will see in §3, amounts to redefining it as the covariance of errors about the implicit ensemble mean.

Summary of §2

Taken together, the freedom in state represented by unobservable variations $\delta x(t)$ and the covariance freedom represented by covariance increments $\delta x(t)AA^T\delta x(t)^T$ will be called the Transparent Variations of the Kalman filter. They are transparent to the filter dynamics in the sense that they can be added to state or covariance without changing anything else. In particular, transparent variations in covariance describe the growth of uncertainty of the unobservable dimensions of a clock system.

Transparent variations in estimation problems can also be exploited in design; they define the freedom in state and covariance that can be used to achieve goals external to the estimation problem. In GPS, there are several such goals, including the steering of GPS time to an external standard such as UTC, the control of the growth of covariance, and the reconciliation of the separate Kalman filters used in different GPS partitions.

Although it is possible as in §1.4, to define a fixed mapping that divides the system into unobservable and observable parts, transparent variations can be used to show the separate behavior of the unobservable parts of a problem *in any state space formulation, even if the state variables are mixtures of observable and unobservable dimensions.* This allows the separate design issues to be recognized and addressed in a full system formulation including all clock states. For purposes of timekeeping, the full system is inherently richer in capability than the observable part of a fixed decomposition.

§3 The Implicit Ensemble Mean

In the earlier master-clock based version of GPS, system time was determined by the master clock. All other clocks

were corrected using estimates of their errors relative to the master. As a result, every corrected clock *represented* the master to within the accuracy of the Kalman filter.

In the present more general formulation with all clocks estimated, we would expect that the corrected clocks will still represent something, perhaps some kind of average of the clocks. We seek a definition of system time for the new system analogous to the master clock in the earlier versions of GPS.

Such a quantity is the implicit ensemble mean of the system. It is defined mathematically in terms of the physical states of all the clocks and the Kalman filter estimates of their bias vectors. Since the physical clock states cannot be available in any one place, the implicit ensemble mean remains implicit. However, the definition leads to certain properties of the implicit ensemble mean in relation to an ideal clock and other properties in relation to individual corrected clocks. In net these results show relationships between an ideal clock and individual corrected clocks that are of great practical value although the implicit ensemble mean itself can never be explicit.

Preview of Results

Any optimal estimate based on measurements is of course at least as good as an open-loop estimate of the same thing, that is, an estimate based on no measurements. Typically, we expect estimates to depend on measurements and to benefit from them in the sense that estimation variances would be smaller with measurements than without. But the implicit ensemble mean is not a typical filter estimate. It is shown that its value is not directly changed by a measurement update.

Moreover, under some common conditions, both the value and the covariance of the implicit ensemble mean are entirely independent of measurements. Specifically, when clock process noise variances remain proportionate from clock to clock (the PPN case of §3.3), both the implicit ensemble mean and its covariance are independent of the number and quality of clock difference measurements. In this case the implicit ensemble mean mean behaves like a single stand-alone clock with reduced process noises. Specifically, its Q's or covariances are related to those of the stand-alone (open-loop) component clocks simply as the inverse of the sum of the inverses.

If process noise statistics for different clocks vary differently with time, yet remain proportional on the average, the implicit ensemble mean still turns out nearly independent of measurements, but not exactly so. There are, however, other examples of non-PPN behavior in which implicit ensemble quality can depend strongly on measurement quality. The most important is the case of clock service interruptions, discussed in §3.4.

In general, each corrected clock (physical clock minus estimated bias) represents the implicit ensemble mean to a high accuracy. Variances of the *representation errors* (differences between corrected clocks and the implicit ensemble mean) will have a steady state or at least stay within a limited range because, as shown in §3.2, they are errors in completely observable subsystems of the system.

Thus the difference between a corrected member clock and the implicit ensemble mean has a bounded variance, so its long-term Allan Variance becomes negligible compared to that of the implicit ensemble mean. Thus the long-term stability of any corrected clock is shown in §3.5 to be that of the implicit ensemble mean.

An important special PPN case is that of n statistically identical clocks. In this case, the Allan Variance (or any other kind of variance) of the implicit ensemble mean is simply n times smaller than that of a single clock, regardless of any measurements.

These conclusions hold although no explicit ensembling equations are implemented; they apply to the results of standard optimal filtering of clock-to-clock difference measurements in a clock system with no master.

§3.1 Definition of the Implicit Ensemble Mean

Definition 3.1: Implicit Ensemble Mean
At any time t, the implicit ensemble mean $\bar{x}^0(t)|X(t)$ of the system of clocks described in §1.1 and 1.2, is defined as the conditional mean of the ideal clock state $x^0(t)$ given the states $x^i(t)$ of all real clocks, given all the available clock difference measurements up to time t, and given no prior information on the ideal clock.

Since the estimation environment is linear and Gaussian, the Kalman filter estimates and covariances at time t form a sufficient statistic for the clock system given all measurements up to time t. Therefore the above definition is equivalent to defining the implicit ensemble mean as the conditional mean of the ideal clock $x^0(t)$ given the states X(t) of all the real clocks and given the Kalman filter estimates $\hat{B}(t)$ of the differences B(t) between real clocks and the ideal clock.

By the definition 1.1-(14) of biases $b^i(t)$, we have for every real clock i the fundamental relation

$$(1)\qquad x^i(t) = x^0(t) + b^i(t) = x^0(t) + \hat{b}^i(t) + e^i(t)$$

where the $e^i(t)$ are the errors in the estimates $\hat{b}^i(t)$ of $b^i(t)$. This can be rewritten in terms of matrices and super-vectors as follows:

$$(2)\qquad X(t) = H^* x^0(t) + B(t) = H^* x^0(t) + \hat{B}(t) + E(t)$$

$$H^* = \begin{bmatrix} I \\ I \\ I \\ I \end{bmatrix};\quad E(t) = \begin{bmatrix} e^1(t) \\ e^2(t) \\ e^3(t) \\ e^4(t) \end{bmatrix};\quad E[E(t)E(t)^T] = \hat{C}(t)$$

where as before the submatrices of H^* are 2x2 identity matrices, E[ ] is an expectation operator, not to be confused with E(t) the vector of clock estimation errors for all the clocks. As before, we illustrate for the case of n = 4 clocks.

At any one time t, if we were given X(t) and $\hat{B}(t)$ and nothing else, (2) allows us to consider the difference $X(t)-\hat{B}(t)$ as one large vector measurement on $x^0(t)$ whose measurement errors E(t) have measurement covariance given by $\hat{C}(t)$ from the Kalman filter:

$$(3)\qquad X(t)-\hat{B}(t) = H^* x^0(t) + E(t)$$

As a result, we can apply a familiar formula from estimation theory to express the implicit ensemble mean as

the conditional mean of $x^{\theta}(t)$ given X(t) and $\hat{B}(t)$ and nothing else:

$$(4)\quad \bar{x}^{\theta}(t)|X(t) = [H^{*T}\hat{C}(t)^{-1}H^{*}]^{-1}H^{*T}\hat{C}(t)^{-1}[X(t)-\hat{B}(t)]$$

Since we will be interested in the error in the implicit ensemble mean relative to an ideal clock, we introduce a shorter notation for this

$$(5)\quad \Delta(t) = \bar{x}^{\theta}(t)|X(t) - x^{\theta}(t) \qquad \text{Def of } \Delta(t)$$

Substituting (3) and (4) into (5) and simplifying yields an equation for the error $\Delta(t)$ in the implicit ensemble mean in terms of filter estimation errors E(t):

$$(6)\quad \Delta(t) = [H^{*T}\hat{C}(t)^{-1}H^{*}]^{-1}H^{*T}\hat{C}(t)^{-1}E(t)$$

This gives a means of evaluating the covariance of the implicit ensemble mean about an ideal clock. Defining

$$(7)\quad C_{\Delta}(t) = E[\Delta(t)\Delta(t)^{T}] \qquad \text{Def of } C_{\Delta}(t)$$

we can apply (6) to obtain, after simplification,

$$(8)\quad C_{\Delta}(t) = [H^{*T}\hat{C}(t)^{-1}H^{*}]^{-1}$$

Although the implicit ensemble mean itself (4) cannot be evaluated in practice for lack of access to X(t), its covariance about an ideal clock is thus readily expressed in terms of the Kalman filter error covariance.

A Removable Singularity

The double inversion in (8) occurs in many formulas involving implicit ensembling. Although such formulas normally need not be evaluated in practice, it is important that they remain well-defined in many cases even though $\hat{C}^{-1}$ may not exist. In particular, if $\hat{C}$ is singular but can be restored to full rank by the addition of a form $H^{*}\,S\,H^{*T}$ for some symmetric semidefinite S, that is, if $\hat{C} + H^{*}\,S\,H^{*T}$ has full rank, then the following identity can be used to remove the singularity:

$$(9)\quad [H^{*T}[C + H^{*}S\,H^{*T}]^{-1}H^{*}]^{-1} = [H^{*T}C^{-1}H^{*}]^{-1} + S$$

This identity is not specific to our H^{*}. It is general, assuming only that H^{*} has independent columns and that the other mentioned matrices and inverses exist. However, when all but C^{-1} exist, (9) supplies a generalized definition of the double inversion in (8):

$$(10)\quad [H^{*T}C^{-1}H^{*}]^{-1} = [H^{*T}[C + H^{*}S\,H^{*T}]^{-1}H^{*}]^{-1} - S$$

Using (10) to generalize the domain of definition of (8) in this way amounts simply to showing the removability of certain singularities.

A proof of (9) is obtainable from a familiar estimation theory formula:

$$[C^{-1} + H^{T}R^{-1}H]^{-1} = C - C\,H^{T}[R + H\,C\,H^{T}]^{-1}H\,C$$

relating the two standard forms for a measurement update to covariance. Premultiplying both sides by H and post-multiplying by its transpose gives

$$H[C^{-1}+H^{T}R^{-1}H]^{-1}H^{T} = HCH^{T} - HCH^{T}[R + HCH^{T}]^{-1}HCH^{T}$$
$$= R[R + HCH^{T}]^{-1}HCH^{T}$$

Inverting both sides (possible assuming H has independent rows) gives:

$$[H[C^{-1} + H^{T}R^{-1}H]^{-1}H^{T}]^{-1} = [HCH^{T}]^{-1}[R + HCH^{T}]R^{-1}$$
$$= [HCH^{T}]^{-1} + R^{-1}$$

which becomes (9) upon substituting C for C^{-1}, S for R^{-1}, and H^{*T} for H.

§3.2 Representation by Corrected Member Clocks

Although the implicit ensemble mean depends on all the physical clocks and is therefore inaccessible as such, an individual corrected clock can be made accessible (with some limitations) to system users. This can be done in many ways, but the simplest is to occasionally deliver to users the estimated bias vectors as determined by the Kalman filter. This means that any user able to receive timing signals from a physical clock can use the estimated biases to adjust to the corrected clock. This is, in essence, the situation in GPS.

This section shows how any individual corrected clock, that is, any physical clock corrected by its estimated bias vector, *represents* the implicit ensemble mean to a very high and computable accuracy.

The difference between an individual corrected clock and the implicit ensemble mean is therefore called the *representation error* for that clock. It is shown to be a weighted sum of the errors in a completely observable subsystem and consequently its covariance either has a steady state or oscillates within a limited range. Thus in a clock system with no master, any individual corrected clock represents the implicit ensemble mean in the same way that it would represent the master clock in a system that has one. The main difference is that the implicit ensemble mean is a much more accurate and reliable standard to be representing.

Dropping the time arguments in 3.1-(4), the implicit ensemble mean can be rewritten:

$$(1)\quad \bar{x}^{\theta}|X = [H^{*T}\hat{C}^{-1}H^{*}]^{-1}H^{*T}\hat{C}^{-1}[X-\hat{B}]$$

whereas the i-th corrected clock (physical clock corrected by estimated bias) is $x^{i} - \hat{b}^{i}$, defined in 1.3-(1) as ω^{i}, and the system of all corrected clocks is therefore denoted by Ω:

$$(2)\quad \omega^{i}(t) = x^{i}(t) - \hat{b}^{i}(t) = x^{\theta}(t) + e^{i}(t) \qquad \text{Def of } \omega^{i}$$
$$\Omega(t) = X(t) - \hat{B}(t) = H^{*}x^{\theta}(t) + E(t) \qquad \text{Def of } \Omega$$

where the first part of each equation is a definition, and the second part follows from the first by 3.1-(1). Thus the implicit ensemble mean is a linear combination of the corrected clocks.

The difference between any corrected clock i and the implicit ensemble mean is called the *representation error* of that clock, denoted by γ^{i}:

$$(3)\quad \gamma^{i}(t) = \omega^{i}(t) - \bar{x}^{\theta}(t)|X(t) \qquad \text{Def of } \gamma^{i}$$
$$= x^{\theta}(t) + e^{i}(t) - [H^{*T}\hat{C}^{-1}H^{*}]^{-1}H^{*T}\hat{C}^{-1}[X(t)-\hat{B}(t)]$$

By (2), $X-\hat{B}$ can also be written $H^{*}x^{\theta} + E$, hence

$$(4)\quad \gamma^{i} = x^{\theta} + e^{i} - [H^{*T}\hat{C}^{-1}H^{*}]^{-1}H^{*T}\hat{C}^{-1}[H^{*}x^{\theta} + E]$$
$$= e^{i} - [H^{*T}\hat{C}^{-1}H^{*}]^{-1}H^{*T}\hat{C}^{-1}E$$

a function purely of error vectors e^j. By substituting the identity

(5) $\quad e^i = [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1} H_* \, e^i$

into (4), we can express γ^i in terms of error vector differences:

(6) $\quad \gamma^i = [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1} [H_* \, e^i - E]$

The rightmost factor in brackets consists entirely of component subvectors of the form $e^i - e^j$ for various j and fixed i. This means that the representation error γ^i is an observable subvector of state.

To see this, we have only to recall from §1.4 that relativizing all clocks to any one clock yields a completely observable subsystem for the n-1 relativized clocks, like the master clock version of GPS.

Thus the covariance for a set of differences $e^j - e^i$ for a fixed i will have a steady state if measurement quality is constant. For a system like GPS with periodic changes in measurement availability, it will oscillate within a narrowly bounded range. In either case, the representation error variance for any clock has no secular growth trends.

In particular for GPS, the first components of vectors γ^i are typically of the order of two nanoseconds (1-sigma).

Observable Component of Covariance

In fact, the variance of the representation error γ^i according to (6) is easily computable from the Kalman filter covariance. The second version of equation (4),

(7) $\quad \gamma^i = e^i - [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1} \, E$

invites rewriting in terms of a super-vector Γ consisting of all the subvectors γ^i:

(8) $\quad \Gamma = E - H_* [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1} \, E$

The extra pre-multiplier H_* in (8) arises only to create n identical copies of the last term in (7), since (8) represents n separate instances of equation (7). Using (8), it becomes possible by standard transformations to evaluate the covariance of the super-vector Γ in terms of that of the Kalman filter error super-vector E:

(9) $\quad C_\Gamma = E[\Gamma(t)\Gamma(t)^T] = E[K \, E(t) E(t)^T K^T]$ where

$K = I - H_* \, [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \, \hat{C}^{-1}$

where E[ ] is the expectation operator and E(t) is the system error vector.

Given the definition of $\hat{C}$ as the expectation of $E(t)E(t)^T$, (9) can be evaluated algebraically as follows:

(10) $\quad C_\Gamma = \hat{C} - H_* \, [H_*^T \, \hat{C}^{-1} H_*]^{-1} \, H_*^T \, \hat{C}^{-1} \, \hat{C}$

$$- \hat{C} \, \hat{C}^{-1} H_* \, [H_*^T \, \hat{C}^{-1} H_*]^{-1} \, H_*^T$$

$$+ H_* [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1} \hat{C} \, \hat{C}^{-1} H_* [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T$$

which simplifies to

(11) $\quad C_\Gamma = \hat{C} - H_* [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T$

which we again recognize as *the* observable component of covariance as defined in §2.2-(12).

Thus (11) shows specifically at any time the actual accuracy with which each corrected clock represents the implicit ensemble mean.

Monitoring Individual Clock Performance

A much needed benefit that results from this is a simple way of monitoring the health and stability of individual clocks. To within the small and computable uncertainty of γ^i, the implicit ensemble mean is equal to $x^i - \hat{b}^i$. So the deviation of any physical clock $x^i(t)$ from the best system standard (the implicit ensemble mean) is represented simply by the estimated bias vector, $\hat{b}^i(t)$ itself.

This gives a better basis for judging clock performance than was available in master-clock versions of GPS, since the implicit ensemble mean is a much more stable standard of comparison. Moreover, the accuracy of the comparison stays within tighter limits. With a master clock, the bias estimate for each clock reflected its deviation from the master, but the accuracy to which it showed this depended on the chain of measurement contacts connecting the two. With an implicitly ensembled system, it no longer matters which clocks a given clock is directly in contact with, only how many; the quality resides equally in all of them.

Implicit Ensemble Weights

The representation errors Γ of corrected member clocks and their covariance C_Γ play important roles at every stage of the analysis of clock systems. One central and perhaps initially surprising result is that the weights in the formula for the implicit ensemble mean depend only on the observable component C_Γ of covariance.

Definition 3.2: Implicit Ensemble Weights

The matrix coefficient in the definition 3.1-(4) of the implicit ensemble mean will be called the implicit ensemble weights:

(12) $\quad W_{IE}(\hat{C}) = [H_*^T \hat{C}^{-1} H_*]^{-1} H_*^T \hat{C}^{-1}$

Theorem 3.2A: Implicit Ensemble Weights

Implicit ensemble weights are independent of transparent variations:

(13) $\quad W_{IE}(C) = W_{IE}(C + H_* \, S \, H_*^T)$

This holds for arbitrary symmetric semidefinite S of the appropriate dimension (in the present case 2x2). The proof is a consequence of the identity 3.1-(9), introduced to show the removability of certain singularities in the double inversion:

(14) $\quad [H_*^T [C + H_* S \, H_*^T]^{-1} H_*]^{-1} = [H_*^T C^{-1} H_*]^{-1} + S$

Post-multiplying both sides by H_*^T,

(15) $\quad [H_*^T [C + H_* S \, H_*^T]^{-1} H_*]^{-1} H_*^T$

$$= [H_*^T C^{-1} H_*]^{-1} [I + H_*^T C^{-1} H_* S] H_*^T$$

$$= [H_*^T C^{-1} H_*]^{-1} H_*^T [I + C^{-1} H_* S \, H_*^T]$$

$$= [H_*^T C^{-1} H_*]^{-1} H_*^T C^{-1} [C + H_* S \, H_*^T]$$

Finally, post-multiplying both sides by $[C+H*S\,H*^T]^{-1}$,

$$(16)\quad [H*^T[C+H*S\,H*^T]^{-1}H*]^{-1}H*^T[C+H*S\,H*^T]^{-1} = [H*^TC^{-1}H*]^{-1}H*^TC^{-1}$$

which is (13) expanded via the definition (12), q.e.d.

As a special case, we can substitute C_Γ as defined by (11) for C in this theorem and $H*^TC^{-1}H*$ for S, concluding that $W_{IE}(\hat{C}) = W_{IE}(C_\Gamma)$, in other words, the implicit ensemble weights depend only on the observable component of covariance, C_Γ.

This result is useful in predicting the relative clock weights. We have already seen that the observable component of covariance, environment permitting, will have a steady state. One of the most useful properties of steady state solutions is that they are independent of initial conditions. Theorem 3.2A thus shows that the implicit ensemble weights eventually become independent of initial conditions. In particular, they eventually recover from the effects of any temporary interruptions, such as clock service interruptions.

Summary of §3.2

§2 began by defining unobservable variations $\delta(t)$ of state and resulting unobservable components $\delta\,\delta^T$ of covariance. It ended by showing, for a covariance C(t) and a complete unobservable variation $\delta(t)$, that there was a unique maximal decomposition of C(t) into two parts, one being unobservable, the other with no unobservable component. On this basis these components were labelled *the* unobservable and observable components of C.

We can now recognize equation (11) of the present section, defining the covariance C_Γ of representation errors Γ, as the observable component C' of C as defined in 2.2-(12). Whereas in §2.2 that component was defined as what was left of covariance after removing the largest unobservable component, we have now proved that each subvector γ^i of Γ is a subvector of a completely observable system in the generally recognized sense.

The form of (11) is also significant in another way. We know by the definition (6) that each corrected clock, $\omega^i = x^i - b^i$, is equal to γ^i plus the implicit ensemble mean. It follows that the error $e^i = \omega^i - x^0$ is equal to the sum of γ^i and the error Δ in the implicit ensemble mean. Equation (11) shows that the corresponding covariance submatrices for γ^i and Δ can likewise be simply summed to give the covariance for e^i. This means that the error Δ in the implicit ensemble mean is uncorrelated with and hence independent (for they are jointly gaussian) of the γ^i. More generally the matrix equation (11) proves the following:

Theorem 3.2B: Independence of $\Delta(t)$ and $\Gamma(t)$

The total error $E(t) = \Omega(t) - H*x^0(t)$ in the corrected clocks is the sum of two statistically independent components: $\Gamma(t)$ and $H*\Delta(t)$, where $\Gamma(t)$ consists of the representation errors $\gamma^i(t)$ for each individual clock and $H*\Delta(t)$ consists of n replicas of the common error $\Delta(t)$ in the implicit ensemble mean.

We have shown that representation errors $\Gamma(t)$ can be expressed in terms of a completely observable subsystem. By 3.1-(8), the other component $H*\Delta$ has covariance:

$$C_{H*\Delta} = H*[H*^TC^{-1}H*]^{-1}H*^T$$

an unobservable component of covariance in the sense defined in §2.2.

§3.3 The PPN Case - Open-Loop Behavior

The last section considered representation errors, that is, differences of corrected clocks from the implicit ensemble mean. Such errors arise from completely observable subsystems and, in a stationary environment, will have a statistical steady state. In all cases they obviously depend on available measurements.

This section considers instead the implicit ensemble mean itself and its behavior which is in many ways independent of measurements. An important special case, the PPN (Proportionate Process Noise) case is described in which the implicit ensemble mean, and its covariance relative to an ideal clock, are entirely independent of the quality and even the existence of measurements.

Intuitively, the idea is very plausible that measurements of clock differences contribute no information on absolute time. One might even hope for a point of view that would make this self-evident, or at least amenable to easy proof. But the problem is not that simple. In fact, the result does not follow unless the clock process noises satisfy certain restrictions. For clock systems that do not meet these conditions, the implicit ensemble mean and covariance are often insensitive to measurements, but *not* independent of them. Later on, we will illustrate a class of extreme but important situations in which the implicit ensemble mean can depend very strongly on measurements.

Meanwhile, the main purpose of this section is to demonstrate sufficient conditions for the implicit ensemble mean to be *entirely* independent of clock difference measurements: This will be the case for any system of independent clocks in which the process noise coefficients for different clocks remain in fixed proportions to each other. This special case is called the *Proportionate Process Noise* (PPN) case. It is defined below.

As a by-product of the analysis, we will begin to appreciate in §3.4 the real value to the ensemble of clock difference measurements in the case of extreme departure from the PPN case following clock service interruptions.

In analyzing the PPN case, it is important to keep in mind that it is a special case, not an assumption. The implicit ensembling properties of a Kalman filter exist for PPN and non-PPN cases alike. The PPN case is of special interest because it yields many simplifying properties, justifying rules of thumb that are intuitively satisfying and useful for predicting the behavior of actual systems.

Definition 3.3: Proportionate Process Noises

Two clocks i and j will be said to have Proportionate Process Noises if there is a scalar constant s relating their process noise coefficients:

$$(1)\quad q_k^i(t) = s\,q_k^j(t), \qquad \text{for } k=1,2 \text{ and for all } t$$

If every pair of clocks in the system have proportionate process noises, then in terms of process noise coefficients, the clocks are all replicas, scaled up or down, of one

original. In other words there exist functions of time $q_1(t)$, $q_2(t)$ and n scalar constants $s_1,\ldots,s_n$ such that

$$(2)\qquad \begin{bmatrix} q_1^i(t) \\ q_2^i(t) \end{bmatrix} = s_i \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix}, \qquad \text{for } i=1,\ldots,n, \text{ all } t$$

For any given interval $(t_k, t_k+\tau)$, a similar relation holds for discrete time process noise covariances: there is $Q(t_k,\tau)$ such that

$$(3)\qquad Q^i(t_k,\tau) = s_i\, Q(t_k,\tau), \qquad \text{for } i=1,\ldots,n$$

where the scalars s_i are fixed for all t,τ and the same as used in (2).

The Open-Loop Version of a Clock System

For a system of clocks with proportionate process noises, we will show that the implicit ensemble mean with any given set of clock-to-clock difference measurements turns out exactly as it would have in the open-loop case, that is, the case of no measurements at all.

As before, we denote by $\hat{C}(t)$ and $\hat{B}(t)$ the covariance matrix and bias estimate vector based on available clock difference measurements up to and including time t. For the open-loop case - no measurements at all - covariance results from process noise alone and is written P(t). The open loop bias estimate arises from pure prediction and thus remains identically zero.

The open-loop covariance for any system of independent clocks propagates as decoupled blocks since there are no measurements to couple them. We will denote individual diagonal blocks by $P^i(t)$:

$$(4)\qquad P(t) = \begin{bmatrix} P^1(t) & 0 & \cdots & 0 \\ 0 & P^2(t) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & P^n(t) \end{bmatrix}$$

For a PPN system, the individual clock covariances $P^i(t)$ are not only independent but also mere replicas of each other, scaled by s_i as their process noises are scaled in (2) or (3). In other words, there is a single 2x2 matrix M(t) such that, for all i, $P^i(t) = s_i M(t)$:

(5):

$$P(t) = \begin{bmatrix} s_1M(t) & 0 & \cdots & 0 \\ 0 & s_2M(t) & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & s_nM(t) \end{bmatrix} = M(t) \begin{bmatrix} s_1I & 0 & \cdots & 0 \\ 0 & s_2I & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & & s_nI \end{bmatrix}$$

Theorem 3.3A: Open-Loop Implicit Ensemble Weights (PPN Case)

For a PPN system of clocks, if for some time t_m, $H*^T P_m^{-1} = H*^T C_m^{-1}$, then for all later times t_k, $k > m$:

$$(6)\qquad H*^T C_k^{-1} = H*^T P_k^{-1}$$

Proof:

The argument is by induction on the Kalman filter update cycle - which we divide into three steps (instead of the usual two): measurement update, pure prediction, and process noise addition. We use the tilde (~) and hat (^) to distinguish values just before and after a measurement update respectively. Values after prediction but before adding process noise are indicated by the absence of any tilde or hat.

Measurement update: $\hat{C}_k^{-1} = \tilde{C}_k^{-1} + H^T R^{-1} H$, $\hat{P}_k^{-1} = \tilde{P}_k^{-1}$

Since $H*$ is a value of an unobservable variation, $H*^T H^T R^{-1} H = 0$. So $H*^T \hat{C}_k^{-1} = H*^T \tilde{C}_k^{-1}$ Thus (6) is preserved under measurement update.

Prediction: $C_k = \Phi\, \hat{C}_{k-1}\, \Phi^T$, $P_k = \Phi\, \hat{P}_{k-1}\, \Phi^T$

Then $H*^T C_k^{-1} - H*^T P_k^{-1} = \phi^{T-1}\left[H*^T \hat{C}_{k-1}^{-1} - H*^T \hat{P}_{k-1}^{-1}\right]\Phi^{-1}$ which is zero if the term in brackets was, thus again preserving the identity (6). This step uses the facts that Φ is block diagonal with ϕ in all diagonal blocks and that $H*$ consists of identity blocks. This allows a kind of commuting of Φ through $H*$ into ϕ, that is, $\Phi\, H* = H*\, \phi$ and $\Phi^{-1} H* = H*\, \phi^{-1}$.

Process noise addition: $\tilde{C}_k = C_k + Q_k$, $\tilde{P}_k = P_k + Q_k$

This is the most complex step and the only one that uses the PPN assumption. If $H*^T C_k^{-1} = H*^T P_k^{-1}$, then

$$(7)\qquad H*^T = H*^T P_k^{-1} C_k = H*^T \begin{bmatrix} s_1 M & & 0 \\ & \ddots & \\ 0 & & s_n M \end{bmatrix} C_k$$

where M is the open-loop 2x2 submatrix assured by the PPN assumption as in (5). We will show (7) is preserved under the addition of process noise to both C_k and P_k:

$$(8)\qquad H*^T \tilde{P}_k^{-1} \tilde{C}_k =$$

$$H*^T \begin{bmatrix} s_1(M+Q) & & 0 \\ & \ddots & \\ 0 & & s_n(M+Q) \end{bmatrix}^{-1} \left[C_k + \begin{bmatrix} s_1 Q & & 0 \\ & \ddots & \\ 0 & & s_n Q \end{bmatrix} \right]$$

$$= H*^T \begin{bmatrix} s_1(M+Q) & & 0 \\ & \ddots & \\ 0 & & s_n(M+Q) \end{bmatrix}^{-1} C_k$$

$$+ H*^T \begin{bmatrix} s_1(M+Q) & & 0 \\ & \ddots & \\ 0 & & s_n(M+Q) \end{bmatrix}^{-1} \begin{bmatrix} s_1 Q & & 0 \\ & \ddots & \\ 0 & & s_n Q \end{bmatrix}$$

$$= (M+Q)^{-1} M\, H*^T \begin{bmatrix} s_1 M & & 0 \\ & \ddots & \\ 0 & & s_n M \end{bmatrix}^{-1} C_k + (M+Q)^{-1} Q\, H*^T$$

$$= (M+Q)^{-1} M \left[H*^T P_k^{-1} C_k\right] + (M+Q)^{-1} Q\, H*^T$$

in which the term in brackets was equal to $H*^T$ by the hypothesis (7), so the whole expression reduces to $(M+Q)^{-1} M\, H*^T + (M+Q)^{-1} Q\, H*^T = H*^T$, q.e.d.

The above repeatedly exploited the fact that a common diagonal block such as M, (M+Q), or their inverses, can be commuted through $H*$ since $H*$ consists of identity blocks.

Recapitulating: the identity $H*^T C^{-1} = H*^T P^{-1}$ is preserved under both measurement and time updates, including, in

the PPN case, the step of adding process noise. This was independent of the specific H and R matrices involved, and therefore of the quality and number of measurements.

Of course, prior to any measurements, the open-loop covariance P and closed-loop covariance C are the same. It follows, since $H*^T C^{-1} = H*^T P^{-1}$ initially, that it remains so, and therefore the weighting matrix in the formula 3.1-(4) for the implicit ensemble mean:

(9) $W_{IE} = (H*^T C^{-1} H*)^{-1} H*^T C^{-1}$ I.E. Weights

can also be determined from the open-loop case:

(10) $W_{IE} = (H*^T P^{-1} H*)^{-1} H*^T P^{-1}$ (PPN case)

Finally, it follows that the covariance of the implicit ensemble mean as given in 3.1-(8):

(11) $C_\Delta = (H*^T C^{-1} H*)^{-1}$

can also be determined from the open-loop case:

(12) $C_\Delta = (H*^T P^{-1} H*)^{-1}$ (PPN case)

For the PPN case, we have so far shown only that the weights and covariances for the implicit ensemble mean are unaffected by measurements. It remains to show that the implicit ensemble mean itself is unaffected. Referring back to 3.1-(4), the implicit ensemble mean vector is

(13) $\bar{x}^0|X = (H*^T C^{-1} H*)^{-1} H*^T C^{-1} (X - \hat{B})$

Since we have already seen that $H*^T C^{-1}$ is unaffected and since $\hat{B} \equiv 0$ for the open-loop case, it will suffice to show:

Theorem 3.3B: Open-Loop I.E.M. (PPN Case):

For a PPN system of clocks, if for some m, $H*^T C_m^{-1} \hat{B}_m = 0$, *then, for all k≥m,* $H*^T C_k^{-1} \hat{B}_k = 0$.

Again, the proof is an induction on the filter update cycle:

Measurement Update: $\hat{C}_k^{-1} = \tilde{C}_k^{-1} + H^T R_k^{-1} H$, and, letting Δz be a measurement residual, $\hat{B}_k = \tilde{B}_k + \tilde{C}_k H^T (R_k + H \tilde{C}_k H^T)^{-1} \Delta z$.

We wish to show $0 = H*^T \hat{C}_k^{-1} \hat{B}_k$. Since $H*^T H^T R^{-1} H = 0$,

(14) $H*^T \hat{C}_k^{-1} \hat{B}_k = H*^T \tilde{C}_k^{-1} \hat{B}_k$
$= H*^T \tilde{C}_k^{-1} \tilde{B}_k + H*^T \tilde{C}_k^{-1} \tilde{C}_k H^T (R_k + H \tilde{C}_k H^T)^{-1} \Delta z$

whose last term vanishes since $H*^T H^T = 0$, leaving the desired identity.

Prediction: $C_k = \Phi \hat{C}_{k-1} \Phi^T$, $B_k = \Phi \hat{B}_{k-1}$.

Hence $H*^T C_k^{-1} B_k = H*^T \Phi^{T-1} \hat{C}_{k-1}^{-1} \Phi^{-1} \Phi \hat{B}_{k-1}$, that is, $= \phi^{T-1} H*^T \hat{C}_{k-1}^{-1} \hat{B}_{k-1}$ which is zero if $H*^T \hat{C}_{k-1}^{-1} \hat{B}_{k-1}$ was.

Process noise addition: $\tilde{C}_k = C_k + Q$, $\tilde{B}_k = B_k$

By the previous theorem,

(15) $H*^T \tilde{C}_k^{-1} \tilde{B}_k = H*^T \tilde{P}_k^{-1} B_k = H*^T (P_k + Q)^{-1} B_k$

$$H*^T \begin{bmatrix} s_1(M+Q) & & 0 \\ & \ddots & \\ 0 & & s_n(M+Q) \end{bmatrix}^{-1} B_k = (M+Q)^{-1} M H*^T \begin{bmatrix} s_1 M & & \\ & \ddots & \\ & & s_n M \end{bmatrix}^{-1} B_k$$

$$= [(M+Q)^{-1} M] [H*^T P_k^{-1} B_k] = [(M+Q)^{-1} M] [H*^T C_k^{-1} B_k]$$

which is zero if its rightmost factor was, q.e.d.

Thus we have shown that $H*^T C^{-1} \hat{B}$ remains permanently zero if it starts out so, and the actual value of the ensemble mean vector is therefore independent of measurements for the PPN case.

Summary of §3.3

An important feature in theorems 3.3A and 3.3B was that the measurement update itself preserved the identities in question *without regard to the PPN assumption.* Yet the measurement update is the only part of the Kalman filter update cycle where the number or quality of measurements are involved at all. The explanation is that although a measurement update cannot affect the present value of $H*^T C^{-1}$, *it can affect a future value by affecting C.*

Yet this cannot happen through the dynamics of prediction only, for then the resulting changes in future values of $H*^T C^{-1}$ due to a present measurement update $H^T R^{-1} H$ would be of one of the forms:

$H*^T H^T R^{-1} H$, $H*^T \Phi^{T-1} H^T R^{-1} H \Phi^{-1}$, $H*^T \Phi^{T-2} H^T R^{-1} H \Phi^{-2}$, ...

all of which are zero by 3.1-(5).

In fact, the proofs of the last two theorems for the PPN case already implied two theorems for the general case, whether PPN applies or not:

Theorem 3.3C: Implicit Ensemble Measurement Update

The implicit ensemble mean and its covariance are unaffected by any measurement update.

Theorem 3.3D: Implicit Ensemble Prediction

Pure prediction (excluding process noise) affects the implicit ensemble mean and its covariance just as it would affect the mean and covariance of a single stand-alone clock. That is, mean and covariance are premultiplied by ϕ, and covariance is also postmultplied by ϕ^T.

In short, whether PPN applies or not, the implicit ensemble mean and its covariance never deviate at all from open-loop propagation rules for a single equivalent clock *except due to the remaining operation of the Kalman filter cycle: process noise addition.* However, in this third operation, not only the implicit ensemble covariance but the implicit ensemble mean itself can change.

We are left with only one means whereby measurement updates can ever affect either the implicit ensemble mean or its covariance: they must change C and thus alter the way in which $H*^T C^{-1}$ and hence the weights will react to subsequent process noises. Even this, we have seen, cannot happen if the process noises for different clocks remain proportionate.

Thus the potential dependency of $H*^T C^{-1}$ (and therefore the ensemble weights and variances) on measurements is both limited and indirect. It is therefore not surprising to find from simulations that even when clock process noises for different clocks are allowed to vary randomly over a sizable range, the implicit ensemble weights and variances typically agree between the open-loop and closed-loop cases very closely. In fact, for numbers typical of GPS

clock and measurement accuracies, the agreement between open-loop and closed-loop for q's varying randomly from 1 to 10 times normal size, is usually 2 to 3 significant digits in the weights, and more in the resulting ensemble variances.

However, this simply shows that such random variations are not the most interesting kind of deviation from the PPN case. A much more important kind of non-PPN behavior is considered in the next section.

§3.4 Clock Spoiling and Reinitialization

Given that the implicit ensemble mean can depend on measurements at all, the question naturally arises under what circumstances, if any, this dependence can be of primary importance.

By way of an answer, this section considers the fundamental problem of clock service interruptions or other losses of time and frequency information. The posssibility of such events is one of the main reasons for wanting an ensemble in the first place. The associated recovery processes of covariance spoiling and clock reinitialization from other clocks, show how the implicit ensemble mean can depend vitally on measurements and their quality and quantity.

A clock reinitialization can follow a maintenance operation or a detected hardware failure or anomaly. In either case, the problem is a one-time discontinuity of unknown and often large size in clock bias or drift or both. Even if nothing else is wrong, the clock covariance requires a large increment, possibly comparable to the initial covariance of a newly activated clock, to reflect the loss of time and frequency information. Such an increase in clock covariance is called clock spoiling.

[A different case, that of a clock anomaly small enough to go undetected and thus uncompensated by spoiling, is not considered here, except to remark that it implies a direct effect on the implicit ensemble mean, typically an n-th part of the change in the affected clock.]

Functionally, the effect on the Kalman filter of covariance spoiling is to help the given clock reacquire time and frequency information from other clocks in the obvious way, from ordinary clock-difference measurements. This is effectively a reinitialization of the clock, implemented automatically by the Kalman filter. The spoiling also makes the information exchange essentially one-way, in that the unspoiled clocks are virtually unaffected by the measurements against the given clock.

[Note: In practice, reinitialization may be partly manual in that a bias or drift state may be crudely reset by a human, leaving only the remaining part of the uncertainty to be resolved by the filter. However, even then the appropriate covariance spoiling is usually so large compared to normal covariances that it can safely be approximated as infinite.]

A covariance increment due to spoiling implies a change in the implicit ensemble weights and hence in the implicit ensemble mean. A vivid and relevant illustration is the case where the covariance spoiling increment for the affected clock is infinite and, prior to the change, all n clocks had identical q's - a PPN case. Thus the implicit ensemble weights prior to the change were all 1/n (times the 2×2 identity matrix). Afterwards the weight for the affected clock is reduced to zero and all the others are increased to 1/(n-1).

The implicit ensemble mean has been shown to be a weighted combination of the corrected clocks ω^i. Taking clock 1 as the affected clock, the implicit ensemble means before and after the change can be expressed as:

$$(1) \qquad IE- = \sum_{i=1}^{n} \frac{1}{n}\,\omega^i \qquad IE+ = \sum_{i=2}^{n} \frac{1}{n-1}\,\omega^i$$

A little algebra shows that the difference amounts to

$$(2) \qquad IE+ - IE- = -\frac{1}{n}\left[\omega^1 - \frac{1}{n-1}\sum_{i=2}^{n}\omega^i\right]$$

$$= \frac{-1}{n-1}\left[\omega^1 - \sum_{i=1}^{n}\frac{1}{n}\,\omega^i\right] = \frac{-1}{n-1}\,\gamma^1$$

In other words, for n identical clocks, *the change in the implicit ensemble mean induced by full (infinite) spoiling of one clock is an (n-1)-th part of the representation error γ of that clock just prior to the spoiling.*

Thus in most cases of clock spoiling in GPS, the implicit ensemble mean is virtually unaffected, for representation errors are usually only a few nanoseconds in bias, and a few nanoseconds per day in drift.

However, the fact that the change in the implicit ensemble mean depends on the representation error of the clock means it can depend enormously on measurement information. Representation errors for a given clock, though reliably small given sufficient measurements, *can become arbitrarily large* during a long period without measurements for that clock.

Thus the result (3) for the case of n identical clocks is really a two-edged sword, showing that the change in the implicit ensemble mean induced by complete spoiling of a single clock will normally be small, but at the same time showing that it can be extremely large if the given clock had been without measurements for a long period.

Earlier, we saw in theorems 3.3C and 3.3D that the implicit ensemble mean and its covariance propagate in a strictly open-loop fashion under measurement update and pure prediction. We concluded that the only way measurements can affect the implicit ensemble mean at all is to change covariance in such a way as to alter the response of the implicit ensemble mean to subsequent process noise, specifically non-PPN process noise.

We have now constructed a blatant example of this type. The availability of measurement feedback on a given clock can have arbitrarily large effects on the change in the implicit ensemble mean induced by subsequent spoiling of that clock.

Serial Clock Service Interruptions

Next consider the case where a series of servicings or anomalies affect the clocks in the system, not all at once but over a period of time. This means that at different times for different clocks, each clock is subject to a one-time memory loss requiring spoiling of its covariance.

Because such memory losses for different clocks occur at different times, the presence of clock-to-clock difference measurements is an enormous advantage: measurements, by continually redistributing time information among the clocks, are responsible for preventing the total loss of time by the system as a whole. This can be understood as follows: After each clock is spoiled, time and frequency is

reacquired from other clocks by means of clock-difference measurements; thus the affected clock resumes its role in the ensemble, helping to sustain it later on when other clocks are spoiled. *If there were no measurements, each clock would be rendered permanently useless once spoiled, and ensemble time would be lost completely once all n clocks had been serviced.*

Thus a series of clock reinitializations provides the most dramatic example of measurement feedback affecting the implicit ensemble mean. Nor is this example artificial. Even if clock anomalies and servicing interruptions are very rare, given a long enough operating time, they will eventually affect all the clocks.

Note: if the interruptions all occurred at the same time (as would be required in the PPN case), there would then be no valid clocks from which to reinitialize, so the ensemble as a whole would lose track of time whether any measurements were available or not. In other words, this vital role of measurements in preserving the ensemble is specific to the non-PPN nature of the problem - in this case to the fact that the extreme covariance increments (large process noises) occur at different times.

This example suggests a general interpretation of the role of clock-to-clock difference measurements in the implicit ensemble: whenever the *relative quality of different clocks changes over time,* clock-to-clock difference measurements allow the ensemble to adapt its dependence on different clocks to take short term advantage of its currently best components. In contrast, the open-loop ensemble over any interval, say from t_1 to t_2, ends up depending only on the net covariance growth $C(t_2)-C(t_1)$ for each clock, regardless of any time varying q's that produced it. Thus whenever relative clock qualities change over time, the existence of measurements will be advantageous to the implicit ensemble.

§3.5 Accuracy/Stability of Implicit Ensembles

The formula 3.3-(12) for the covariance of the implicit ensemble error has a simple interpretation. Since $H*$ consists of a column of identity submatrices, $H*^T C^{-1} H*$ is simply the sum of all the 2x2 blocks in C^{-1}. In the PPN case, C^{-1} can be replaced by P^{-1}, where P is the open-loop covariance. Since P is block diagonal, we end up with the sum of the n diagonal blocks representing the inverses of individual clock open-loop covariances. Thus the open-loop implicit ensemble has a covariance that is simply the inverse of the sum of the inverses of the individual open-loop clock covariances P^i.

$$(1) \qquad C_\Delta(t) = \left[\sum_{i=1}^{n} P^i(t)^{-1} \right]^{-1} \qquad \text{(PPN case)}$$

Moreover, in the PPN case, each open-loop clock covariance P^i is a scalar multiple s_i M(t) of one matrix, so even the matrix inversions drop out:

$$(2) \qquad C_\Delta(t) = \frac{M(t)}{1/s_1 + 1/s_2 + \ldots + 1/s_n} \qquad \text{(PPN case)}$$

Still more specifically, in the case where the clocks all have the same process noises, we would have all $P^i(t)$ equal, and

$$(3) \qquad C_\Delta(t) = \frac{1}{n} P^i(t) \qquad \text{(n identical clocks)}$$

Thus is the same as to the covariance of a single clock in which the q's are all reduced by a factor of n.

It is also useful to interpret the performance of the implicit ensemble mean in terms of stability, usually measured by the Allan Variance. The Allan Variance of a stationary random process x(t) is most simply defined as a function of a time interval τ as follows:

$$(4) \qquad AV(\tau) = \frac{1}{2\tau^2} E\left[x(t) - 2x(t+\tau) + x(t+2\tau)\right]^2$$

where the expectation operator $E[\cdot]$ applies over all time as well as all possible instances of the random process.

Allan Variances under this definition can be evaluated as a simple function of the q's:

$$(5) \qquad AV(\tau) = \frac{1}{\tau} q_1 + \tau q_2$$

Clearly, in the case of n identical clocks, the fact that the implicit ensemble mean behaves as a single clock with its variances scaled as in (2), means that its Allan Variance is similarly so scaled. Although an all Cesium version of GPS could actually satisfy these assumptions, current GPS of course does not exactly meet the PPN requirement since it has some Rubidium clocks with a third (drift rate) estimated state. Nonetheless, for many purposes, GPS covariances behave as if the PPN assumption were satisfied.

We showed earlier that individual corrected clocks can be expressed as the sum of the implicit ensemble mean and their representation errors, of which the latter have constant or at least bounded variances. Given any such bound K, for large τ the three terms in the Allan Variance (4) for the representation error become independent, adding up to at most $3K/\tau^2$. Thus for large τ the Allan Variance of an individual corrected clock exceeds that of the implicit ensemble mean by at most $3K/\tau^2$. Since that of the implicit ensemble mean itself grows with τ, this added increment quickly becomes negligible.

Since variances K are of the order of the square of a few nanoseconds, this addition to the Allan Variance of the implicit ensemble mean already becomes negligible for modest τ, of the order of a day. For small τ, the differences from the implicit ensemble mean, being errors in observable subvectors, become highly correlated and thus tend to cancel, giving an increment much smaller than $3K/\tau^2$. For intermediate sized τ, the situation is complicated not only by correlations over time but also by the GPS upload schedule, under which estimated errors propagate open-loop for up to 24 hours between updates. Apart from the discrete time effects, simulations show Allan Variances smoothly varying from the small τ case to the large.

The bottom line for stability, however, is long-term Allan Variance, and for this case we have seen that the stability of any corrected member clock, must approach that of the implicit ensemble mean.

§4 Non-Clock States

For brevity of notation, the equations of the preceding sections used state vectors X(t) consisting entirely of clock states. But the results apply to systems with non-clock

states as well, subject to two key assumptions: (a) the clock states and non-clock states are completely decoupled in the dynamic models, and (b) although measurements may depend on both clock and non-clock states, their clock dependence is through clock-to-clock differences only.

The generalized clock system state vector X(t) can be partitioned into n subvectors $x^i(t)$ for clock states as before, plus an added n+1st subvector $x^{n+1}(t)$ of arbitrary dimension m, comprising all the non-clock states. Based on assumption (a) above, the state transition matrix for the non-clock states which we will call $\chi(\tau)$ must be mxm only, and the total transition matrix $\Phi(\tau)$ for the full state vector X(t) becomes (illustrating for the case n = 4):

$$(1)\qquad \Phi(\tau) = \begin{bmatrix} \phi(\tau) & 0 & 0 & 0 & 0 \\ 0 & \phi(\tau) & 0 & 0 & 0 \\ 0 & 0 & \phi(\tau) & 0 & 0 \\ 0 & 0 & 0 & \phi(\tau) & 0 \\ 0 & 0 & 0 & 0 & \chi(\tau) \end{bmatrix}$$

where all submatrices are 2x2 except those in the last row (and column) which are m rows (columns) deep. Thus $\Phi(\tau)$ remains block diagonal as before, but only the first n diagonal blocks are identical, the n+1st is arbitrary and possibly much larger in dimension than the first n.

Based on assumption (b) above, the measurement coefficient matrices are similarly generalized from 1.2-(1) to include an arbitrary n+1st subvector of dimension m:

$$(2)\qquad H=[h^1\ 0\ 0\ h^2\ 0\ 0\ h^3\ 0\ 0\ h^4\ 0\ 0\ h^{n+1}]$$

where for some $i\neq j$, $1\le i,j\le n$, $h^i=1$ and $h^j=-1$ and all other $h^k= 0$ for $k\le n$, and where h^{n+1} is an arbitrary m-vector. The case of a measurement with no clock dependence at all ($h^k= 0$ for all $k\le n$) is also permitted.

The matrix H* is then generalized by appending a zero submatrix, this time of dimension mx2:

$$(3)\qquad H^* = \begin{bmatrix} I \\ I \\ I \\ I \\ 0 \end{bmatrix}$$

With these generalizations of H, H*, and Φ, the identities

$$(4)\qquad \begin{aligned} &H\,H^* = 0 \\ &H\,\Phi^k H^* = 0\ , \quad \text{arbitrary } k \ge 0 \end{aligned}$$

remain valid, and as a result all the proofs in §2 go through as before. It is easy to see as well that the "commuting through" identities relied on in the PPN parts of §3 are also preserved:

$$(5)\qquad \begin{aligned} &H^{*T}\Phi = \phi\, H^{*T} \\ &H^{*T}\Phi^{-1} = \phi^{-1} H^{*T} \end{aligned}$$

With these generalizations, all the proofs of sections 2 and 3 remain valid.

§5. Covariance Reduction

In a clock system such as GPS, the covariances of an orthodox Kalman filter with no special modifications will slowly grow without bound in certain respects due to the unobservable dimensions of the problem. In a sense this is simply realistic, reflecting the fact that GPS as a whole, like any finite ensemble of clocks, will gradually drift away from any absolute time standard. For most ordinary purposes involving the short or medium-term behavior of the GPS Kalman filter, such covariance growth is harmless.

However, over very long enough periods of operation, there is a risk of numerical ill-conditioning as unobservable variance components swamp observable ones. In addition, human users prefer covariances they can interpret in terms of observable errors.

The discussion of transparent variations in §2 showed a direct way to avoid these problems. We can simply remove the unobservable component of covariance, that is replace covariance C by its observable component C' as defined by 2.2-(12) or equivalently C_Γ as defined by 3.2-(11). Such replacement corresponds to a transparent variation and thus has no effect on gains or filter responses, hence none on estimates or corrected clocks. Such replacement can be repeated as often as needed or desired.

This approach preserves the optimality properties of the Kalman filter, assuring conditional means (minimum variance estimates) of all clock errors given the data. In the PPN case, as shown in §3.5, it also assures that system stability in the Allan variance sense is improved by the same factor (the effective number of clocks) that applies to estimation variances.

However, for non PPN systems, it is not clear what the relationship is between optimality and stability of clock estimates. Meanwhile, for PPN systems, the direct removal of the unobservable part of covariance is not the only method with optimal properties.

§5.1 Transparent Pseudo-Measurements and PPN Systems

As early as 1985, thoughts of using the Kalman filter to ensemble the GPS clocks included the idea of limiting covariance growth by means of a "pseudo-measurement." This means a covariance-reducing operation of the form used for a measurement update, but with no corresponding change to estimates. §2 shows that this idea includes the transparent changes to covariance among other things, depending on the form of the pseudo-measurement.

Since covariance growth results from process noises that are distributed over time, any pseudo-measurement would need to be repeated over time. This suggests that for simplicity the pseudo-measurement definition at any one instant should be a single (scalar) combination of states. Since the clock problem inherently has two unobservable dimensions, equivalent to a bias and drift, the only scalar combination type that could resolve both unobservables over time is a combination of biases.

The question then becomes whether any particular linear combination of *bias* states is preferred, such as a simple unweighted sum of the biases. The following argument shows that there is a unique combination of bias states for which, in the PPN case, the resulting change can be shown to be a transparent variation of covariance.

In terms of the clock covariance C, the desired form for the pseudo-measurement covariance adjustment is

$$(1)\qquad \Delta C = CH^T[R + HCH^T]^{-1}HC \qquad \text{where}$$

$$(2)\qquad H = [h_1\ 0\ h_2\ 0\ h_3\ 0\ \ldots\ h_n\ 0]$$

It is shown in §2.2 that a covariance change of the form (1) is a transparent variation if and only if $C\,H^T$ is a transparent variation of state, i.e., for some 2×1 matrix Y,

$$(3) \qquad C\,H^T = H^*\,Y, \qquad \text{hence} \quad H^T = C^{-1}H^*\,Y$$

Since the H required by (2) involves biases only, the first question is whether any H-vector can satisfy (3) while involving biases only, i.e., having zeroes in every even-numbered component. The columns of the matrix $C^{-1}H^*$ have in general both odd and even components.

However, in the PPN case, theorem 3.3A assures that $C^{-1}H^*$ has a special structure: all n of its 2×2 submatrices are scalar multiples of a common 2×2 form. Thus, all the 2×1 matrix Y has to do is map the common 2×2 form into the special 2×1 form $(x,0)^T$, and H will end up consisting entirely of odd-numbered elements. This shows that (2) and (3) can be satisfied for some H.

To determine the specific nature of the required H, we consider a rearrangement of state. Place all the bias states first, all the drift states last. In this case H*, which is fundamentally a replicator, defined by $X(t) = H^*\,x^0(t) + B(t)$ takes a new form, as does covariance itself:

$$(4) \qquad H^* = \begin{bmatrix} u & 0 \\ 0 & u \end{bmatrix} \qquad C = \begin{bmatrix} B & P \\ P^T & D \end{bmatrix} \qquad C^{-1} = \begin{bmatrix} U & V \\ V^T & W \end{bmatrix}$$

where u is a column vector of n ones, B is the n×n covariance of the bias states alone, D the n×n covariance of the drift states alone, P the n×n covariance of bias states with drift states, and U, V, and W are similar partitions of C^{-1}. By definition of inversion, $C\,C^{-1} = I$, this implies

$$(5) \qquad \begin{aligned} B\,U + P\,V^T &= I & B\,V + P\,W &= 0 \\ P^T U + D\,V^T &= 0 & P^T V + D\,W &= I \end{aligned}$$

We can safely assume B and D are invertible, so the off-diagonal parts of (5) give $V=-B^{-1}PW$, $V^T=-D^{-1}P^TU$. Substituting these into the diagonal parts of (5),

$$(6) \qquad \begin{aligned} BU - P[D^{-1}P^TU] &= I, & \text{i.e,} \quad [B-P\,D^{-1}P^T]U &= I \\ -P^TB^{-1}P\,W + DW &= I, & \text{i.e.,} \quad [D-P^TB^{-1}P\,]W &= I \end{aligned}$$

$$(7) \qquad \begin{aligned} U &= [B-P\,D^{-1}P^T]^{-1} & V &= -B^{-1}P\,[D-P^TB^{-1}P\,]^{-1} \\ W &= [D-P^TB^{-1}P\,]^{-1} & V^T &= -D^{-1}P^T[B-P\,D^{-1}P^T]^{-1} \end{aligned}$$

The requirement that $C^{-1}H^*\,Y = H$ with only bias elements means then that its last n elements are zero.

$$(8) \qquad \begin{bmatrix} U & V \\ V^T & W \end{bmatrix} H^*Y = \begin{bmatrix} U\,u & V\,u \\ V^Tu & W\,u \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} Y_1U\,u + Y_2V\,u \\ Y_1V^Tu + Y_2W\,u \end{bmatrix} = \begin{bmatrix} h \\ 0 \end{bmatrix}$$

where the elements of the n-vector h are the odd elements $h_1,h_2,\ldots,h_n$ of the originally sought 2n-vector H. The required Y_1 and Y_2 to nullify the lower vector can exist if and only if V^Tu and W u are parallel.

In the PPN case, it was shown that $C^{-1}H^*$ can be based on the open-loop version of C. In terms of the original ordering of states (bias, drift, bias, drift, etc.), the open-loop covariance is block diagonal, with all n 2×2 blocks on the main diagonal proportional to a common 2×2 shape. In terms of equation (8) with the present rearrangement of states and consequent repartitioning of H* and C, this means that the n×n partitions B, P, and D of C are truly diagonal and proportional to a common n×n diagonal matrix, which we may as well identify with B, and similarly U, V, and W are all proportional to B^{-1}.

This means that V^Tu and W u are indeed parallel, in fact each reflects the proportions of the diagonals of B^{-1}. So the necessary Y_1 and Y_2 exist to give the desired form for H. Moreover, the resulting vector h is also parallel to W u, V u, V^Tu and in particular to $B^{-1}u$.

Summary of §5:

A pseudo-measurement has been formulated that is a weighted combination of the bias states of all the clocks, for which the effect on covariance is equivalent, in the PPN case, to a transparent variation. Based on convenience and empirical demonstrations of stability, this is the approach currently in use in GPS.

For general non-PPN environments, neither this nor any other single scalar pseudo-measurement is both sufficient to prevent unobservable covariance growth and also purely transparent, although this can easily be done with any of various pairs of pseudo-measurements.

Although the purely transparent approach is evidently optimal in the estimation theoretic sense of propagating conditional means of all clock errors, and we saw in §3.3 how estimation improvements directly correspond to Allan Variance stability improvements for the PPN case, it is not yet known what relationship filter optimality bears to short or long term stability in the non PPN case. Thus no claim of full optimality is made for non PPN systems.

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